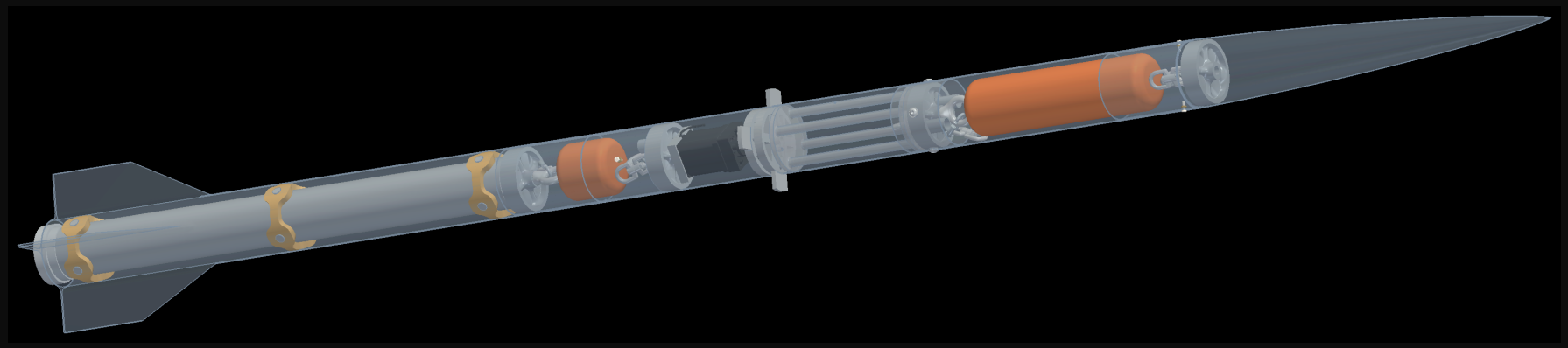


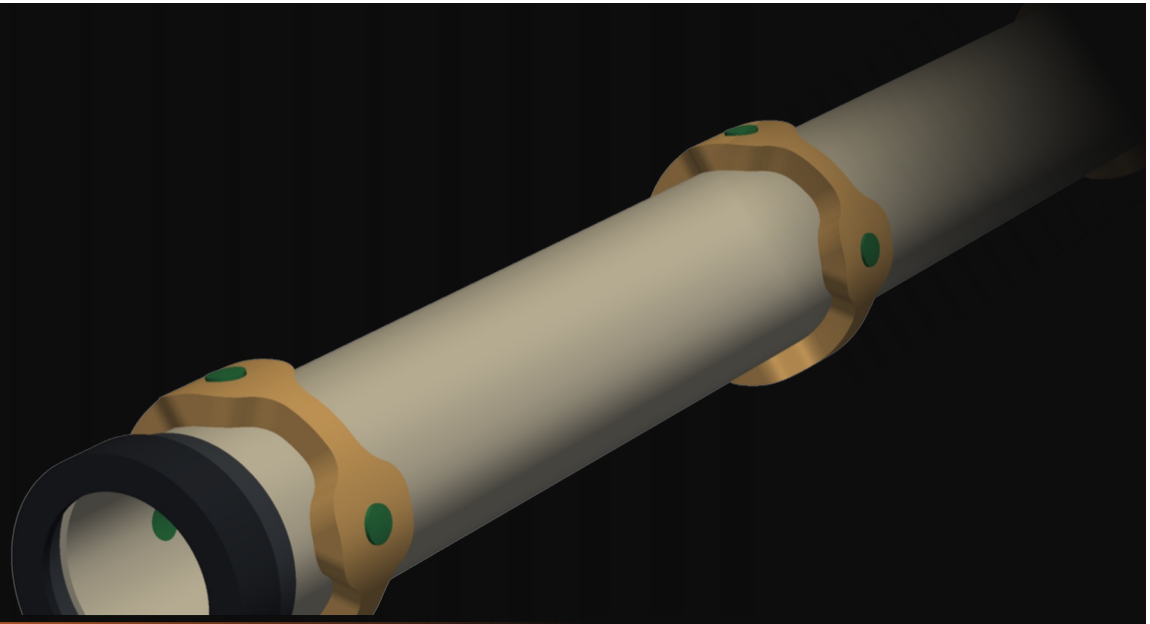
Preliminary Design Review

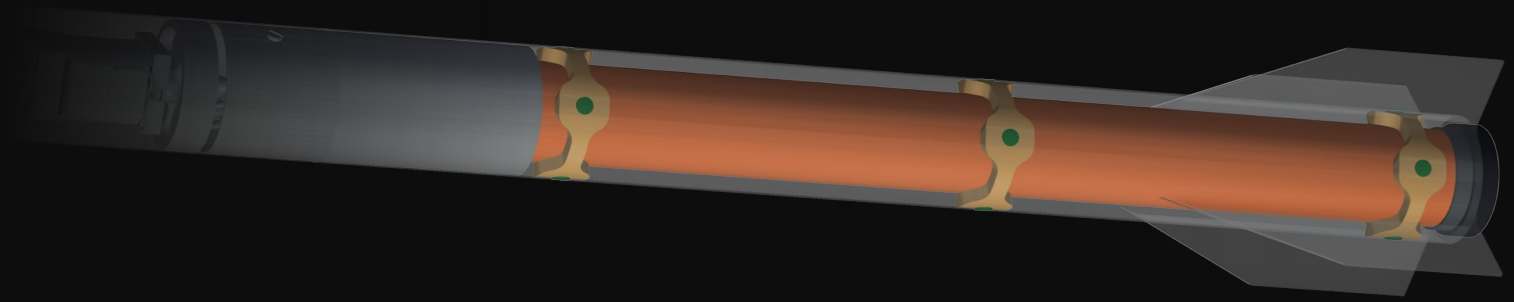
Group 2 · New Member Project 2026

Thrust Structure · **Recovery** · **Airbrakes** · **GNC** · **Aerodynamics**

Andrew McGuigan · David Chertok · Cooper Boris · Helen Garcia

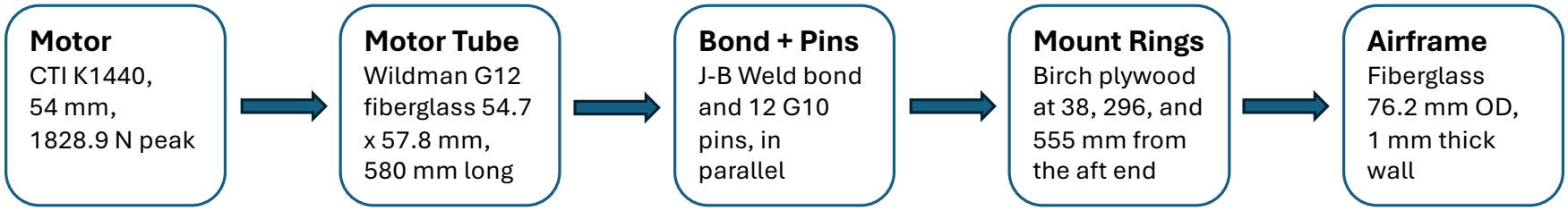
Thrust Structure





Thrust Structure – Role and Load Path

The thrust structure holds the motor on the rocket's centerline and carries its thrust into the airframe



Key Design Rule: The bond and the pins are two independent load paths, and each one is sized to carry the full peak thrust by itself

Thrust Structure – Requirements

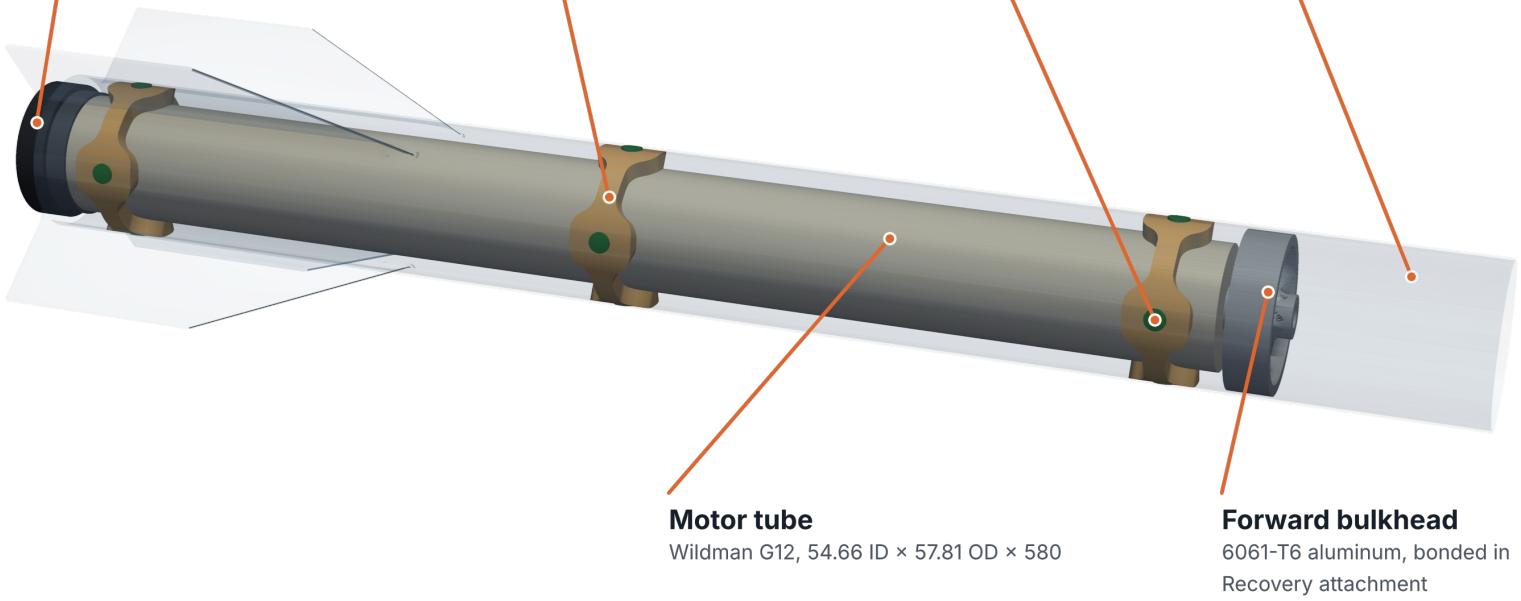
FAR-OUT MANDATORY

Code	Requirement	How it applies	Status
FO.2.8	Use worst-case thrust data	Designs to the 1,828.9 N peak	Met: all checks
FO.2.5	Thrust-to-weight above 6	Liftoff mass ≤ 24.4 kg	Met: 6.98 kg, TWR of 21
FO.8.11	Total impulse ≤ 40,960 N·s	K1440: 2,372 N·s	Met

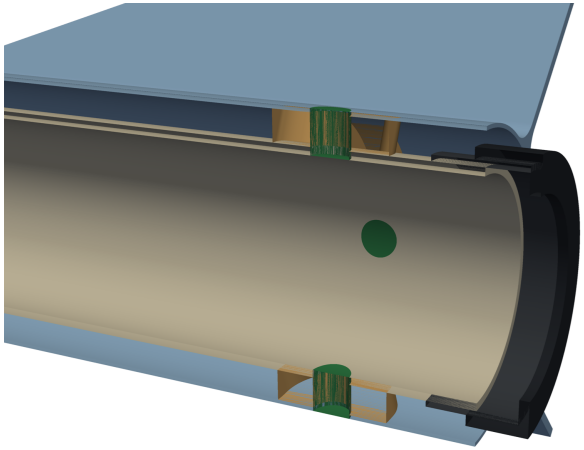
DERIVED FOR THE THRUST STRUCTURE

Code	Requirement	Target	Status
TS.1	Carry peak thrust	1,828.9 N at FOS ≥ 2.0	Met: lowest FOS of 2.55
TS.2	Accept the motor	K1440, 54 x 572 mm	Met: 54.7 mm bore, 580 mm tube
TS.3	Hold motor on centerline	Spaced supports along tube	Met: 3 rings over 517 mm
TS.4	Survive loss of one load path	Bond or pins alone, FOS ≥ 2.0	Met: bond/pins FOS of 8.2/5.3
TS.5	Retain motor on recovery	Motor cannot slide out aft	Met: Aero Pack RA54P retainer

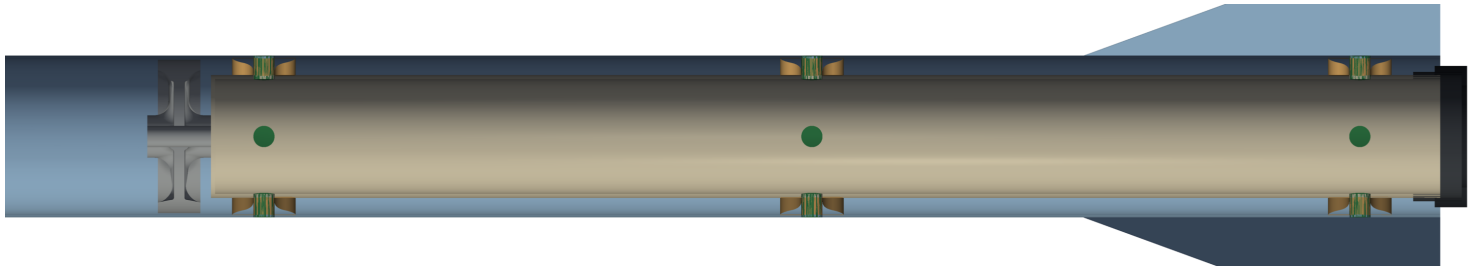
Thrust Structure – CAD Overview



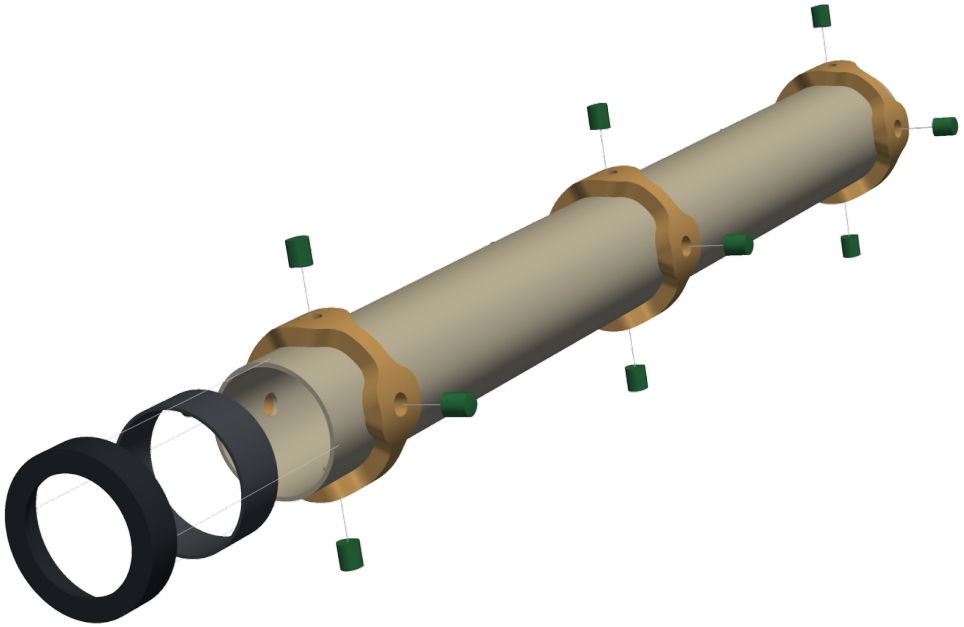
Thrust Structure – Components



Part	Size (mm)	Quantity
Motor Tube	54.66 ID x 57.81 OD x 580	1
Centering Ring	58 ID x 74.2 OD x 30	3
Pin	⌀10 x 11.6, ends contoured	12
Retainer	⌀66.6 cap, 25.4 long	1
Lower Airframe	74.2 ID x 76.2 OD x 690	1
Pin Stations	37.9, 296.4, 555.0 from aft end	4 each



Thrust Structure – Exploded View



Skeletal Ring
42% less material than a solid ring of the same size

Bonded Faces
3,103 mm^2 on the tube and 4,132 mm^2 on the airframe per ring

Four Pins Per Ring
10 mm G10 rod, cut to length and sanded to the curve

Motor Retainer
Aero Pack RA54P ring bonded to the tube, screw-on cap

Thrust Structure – Design Decisions

Pin Shear FOS

0.4 → 25

Plywood pins to G10 rod

Plywood failed in shear along its plies and was crushed by the 1 mm airframe wall. The 10 mm G10 fiberglass rod passes every pin check, for an increase of about 12 g.

Epoxy Temp Limit

60 → 260 °C

J-B Weld on every bond

Worst case, the bond next to the motor passes 60 °C, where standard epoxies lose most of their strength, 1 s into the burn. J-B Weld is rated to 260 °C.

Unsupported Tube Span

530 → 259 mm

Third centering ring

Halves the span between supports and adds 4 pins. Aft pins moved to 37.9 mm, raising tube tear-out FOS from 20 to 34, for an increase of about 26 g.

Thrust Structure – Thermal Check

Bond-line temperature through the 1.57 mm motor tube wall, with the motor case in direct contact and held at its 200 °C certification limit:

1.1 s

Bond line passes 60 °C, where standard epoxies lose most of their strength

84 °C

Bond line at burnout (1.7 s): past standard epoxy, well under J-B Weld's 260 °C

200 °C

Bond line at 60 s: still under J-B Weld's 260 °C rating

Plan: Use the J-B Weld for every bond near the motor (includes retainer, end cap, bulkhead, and both faces of each centering ring)

Adhesive Limits

Standard Epoxy: 20 MPa at 25 °C, only 4.8 MPa at 60 °C

J-B Weld: 260 °C continuous; 9.43 MPa shear after 3 hours at 204 °C

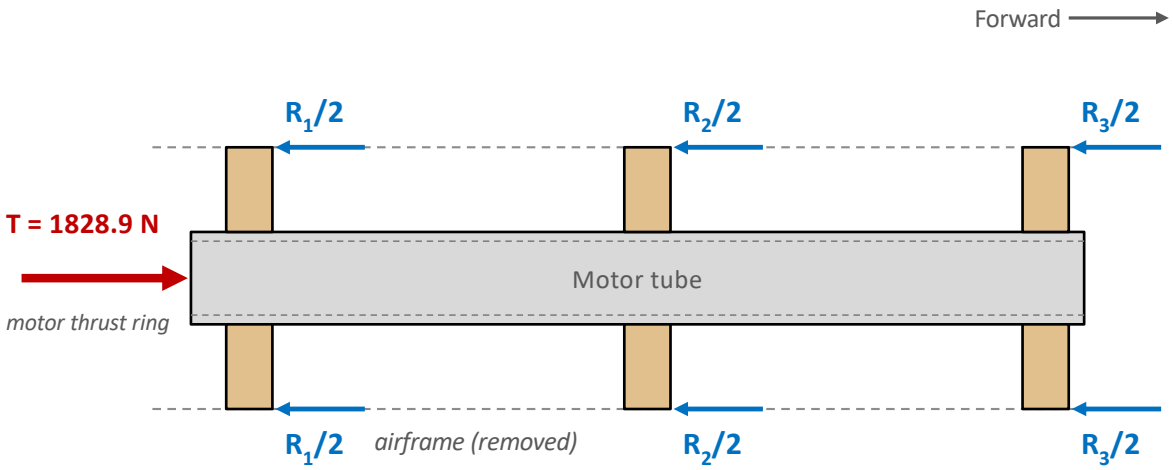
Retainer Bond: J-B Weld, FOS of 58 at an assumed 50 g deployment

Thrust Structure – Materials

Part	Material	Allowable used	Why?
Motor tube	Wildman G12 fiberglass	241 MPa bearing and compression	Matches the airframe; strong in bearing at pin holes
Centering rings	Aircraft-grade birch plywood	9.5 MPa panel shear	Very light and easy to cut
Pins	G10/FR4 rod, ⌀10 mm	148 MPa shear, 345 MPa bending	Light; strong across its layers; stock part
Motor retainer	6061-T6 aluminum (Aero Pack RA54P)	276 MPa yield, 207 MPa shear	Fits the 2.92-inch airframe; tool-free motor changes
Adhesive	J-B Weld Original, for whole thrust structure	4.83 MPa shear (≈ 2/3 of 7.17 MPa, for heat)	Rated to 260 °C continuous; one adhesive for the whole build

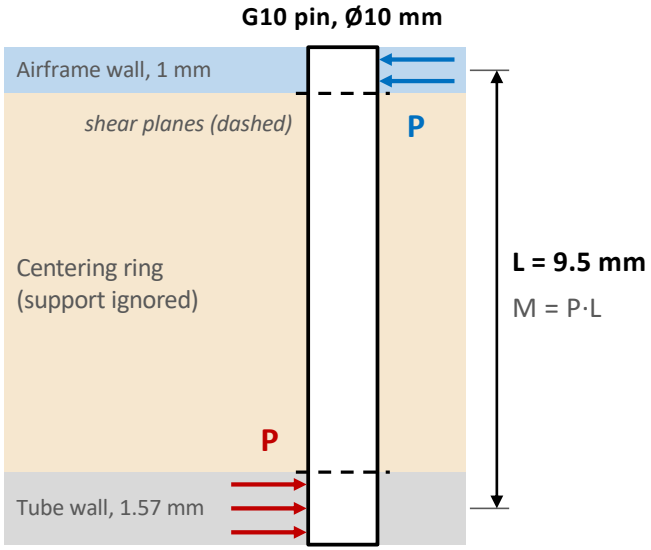
Thrust Structure – Free Body Diagram

(a) Motor tube and centering rings as one free body



$\Sigma F = 0: \quad T = R_1 + R_2 + R_3$

(b) One pin, worst case

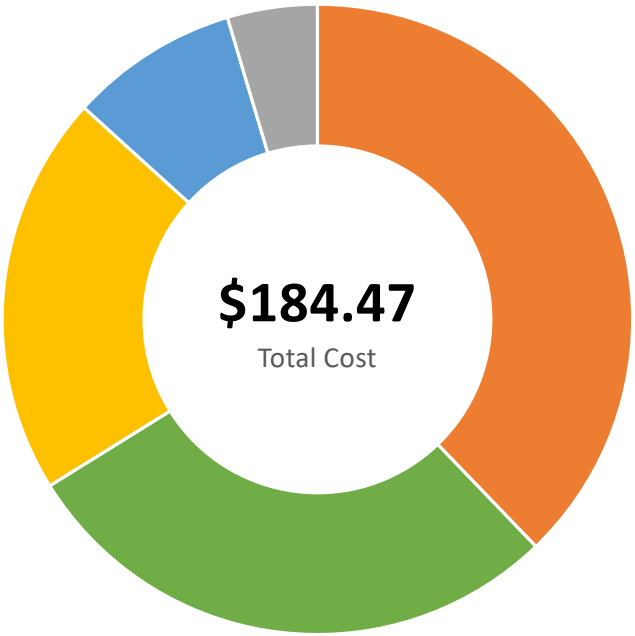


P = T / 4 = 457 N per pin

Thrust Structure – Stress Analysis

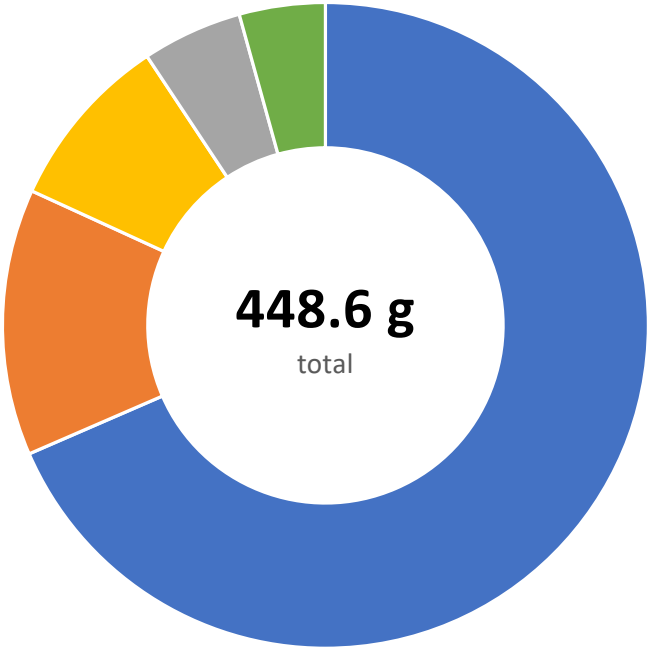
Check	Applied	Allowable	FOS
Bearing, 1 mm airframe	45.7 MPa	241 MPa	5.3
Pin bending (no ring support)	44.3 MPa	345 MPa	7.8
Bond, ring inner face	0.59 MPa	4.83 MPa	8.2
Motor tube bearing, 1.57 mm wall	29.1 MPa	241 MPa	8.3
Ring shear through plywood	0.59 MPa	9.5 MPa	16.1
Pin shear, G10	5.83 MPa	148 MPa	25.4
Airframe compression	7.75 MPa	241 MPa	31.1
Motor tube shear-out, aft holes	4.41 MPa	148 MPa	33.6
Airframe Euler buckling	**1,829 N**	**4,669 N**	**2.55**

Thrust Structure – Cost Budget



	Item	Source	Cost	%
	Birch plywood, 6 mm, 2 × 4 ft (3 rings)	Aircraft Spruce 02-20850	$69.75	38%
	G12 motor tube, 30 in, cut to 580 mm	Wildman G12-2.1	$52.27	28%
	Motor retainer, Aero Pack RA54P	Apogee Components	$37.97	20%
	J-B Weld Original x2	J-B Weld 8265-S	$15.98	9%
	G10/FR4 rod, Ø10 mm × 1 ft (12 pins)	McMaster-Carr	$8.50	5%
	Total		**$184.47**	100%

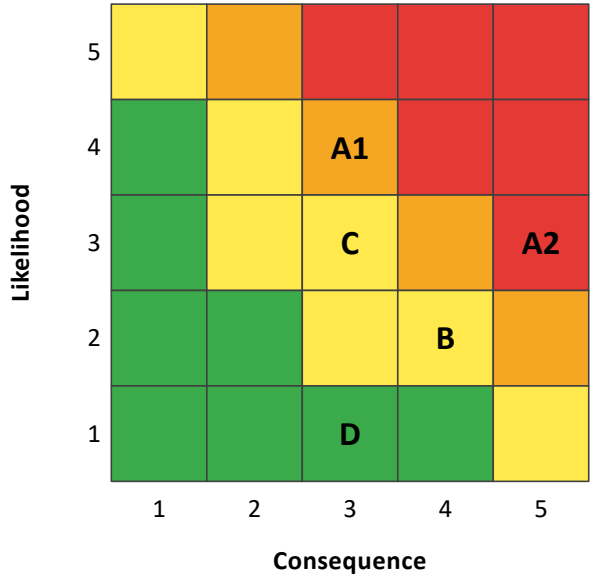
Thrust Structure – Mass Budget

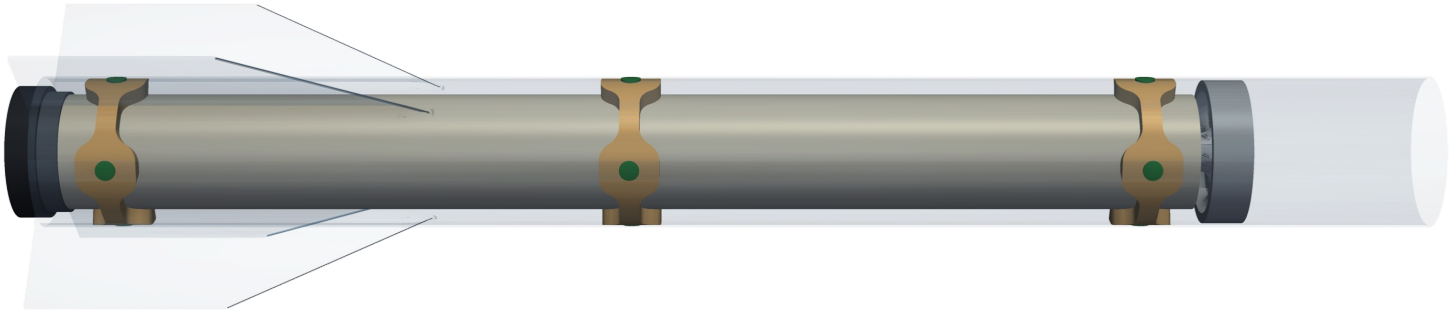


	Component	Basis	Mass (g)	%
	Motor tube + end cap	Wildman G12, 1.85 g/cm³	307.3	68%
	Centering rings ×3	Birch plywood, 0.68 g/cm³	59.9	13%
	Motor retainer, Aero Pack RA54P	Aero Pack published weight	39.7	9%
	J-B Weld bonds (estimate)	Weigh before/after bonding	22.5	5%
	G10 pins ×12	G10/FR4 rod, 1.85 g/cm³	19.3	4%
	Total		**448.6**	100%

Thrust Structure – Risk Assessment

ID	Failure mode → effect	L×C	Score	Action → after
A1	Ring bonds weaken from motor heat → pins still carry thrust; rebond	4×3	**12** High	Hot bond test, wk 5 → **6**
A2	Retainer bond fails from heat → motor case drops out at deployment	3×5	**15** Critical	Hot pull test of retainer bond, wk 5→ **5**
B	Pin holes tear out of fiberglass tubes → rework tubes	2×4	**8** Medium	Drill with backing; FOS 5.3. Accept
C	3D ring contour hard to cut → schedule slip	3×3	**9** Medium	Test cuts, wk 4; fallback flat profile
D	Motor pushes on 2 mm end cap → cap damage	1×3	**3** Low	Rests on aluminum bulkhead. Use as is





Thrust Structure – Next Steps

Estimating six weeks from PDR to CDR

Week 1
- PDR Feedback
- Order parts ($184)
- Begin updating CAD

Week 2
- All part and assembly drawings
- All diagrams
- All plots

Week 3
- FEA of skeletal ring
- Parts arrive

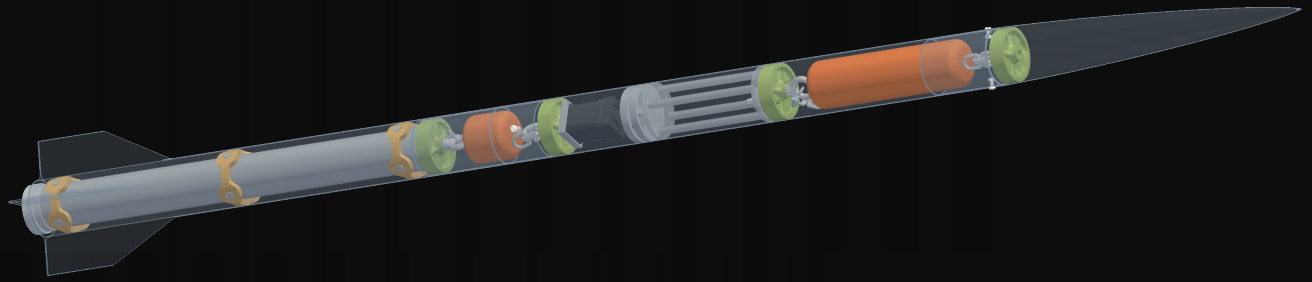
Week 4
- All fit checks with ordered parts
- Test cuts with wood centering rings

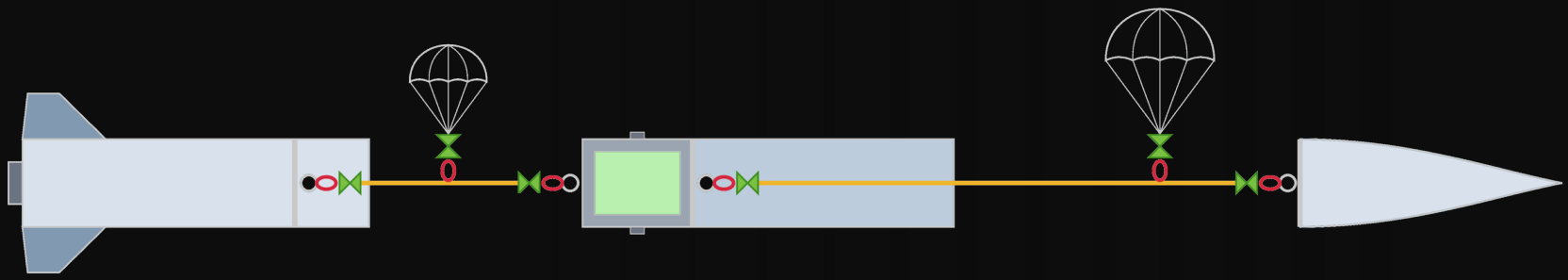
Week 5
- J-B Weld bond testing
- Update and finalize all design choices

Week 6
- Finalize analysis and masses
- Create CDR slides
- Presentation rehearsals

Milestones: PDR (start of week 1) – Parts in hand (end of week 3) – CDR (end of week 6)

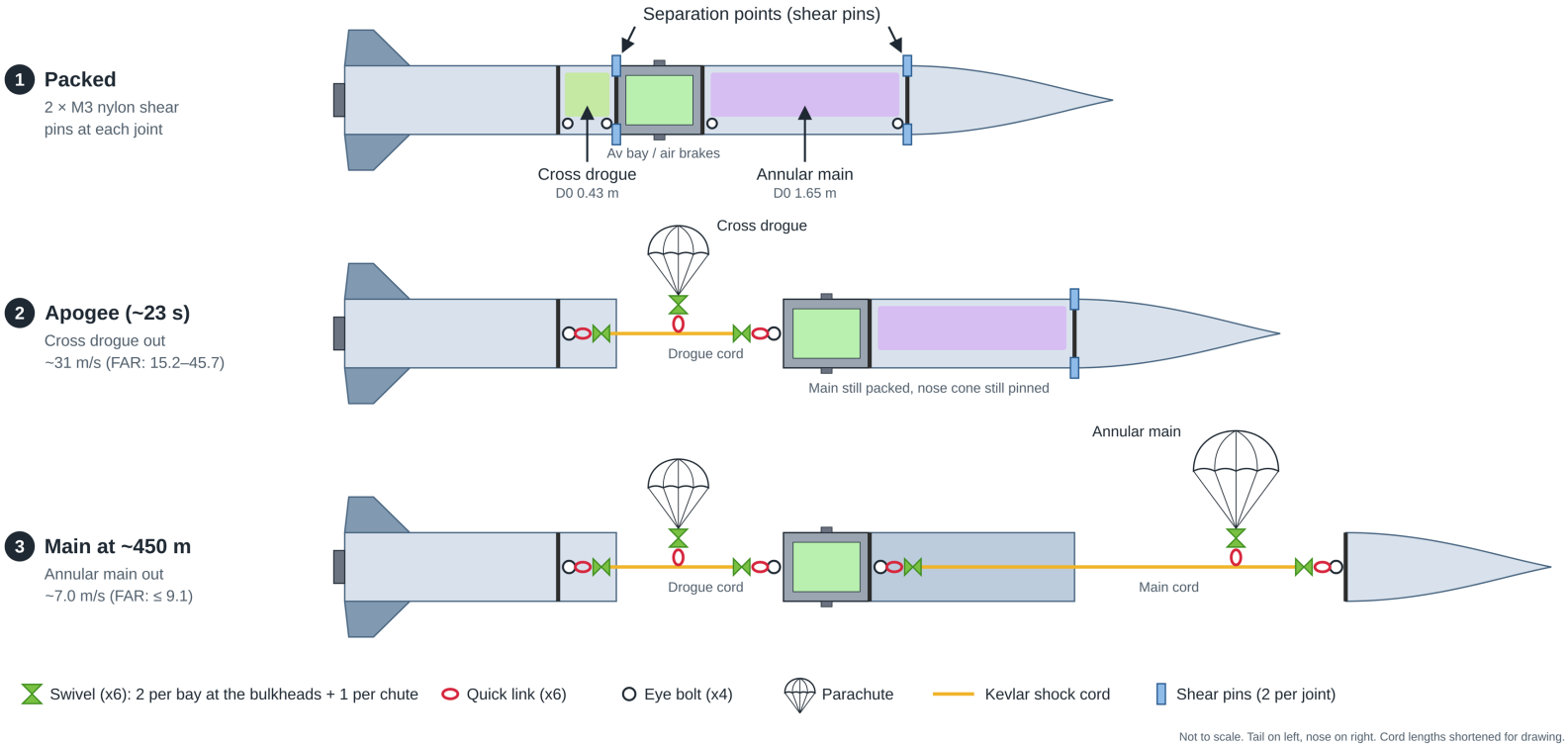
Recovery





Recovery – Dual Deploy Sequence

Dual deploy brings the rocket down in two stages: a small drogue at apogee, then the main near the ground.



Recovery – Requirements

FAR-OUT MANDATORY: DEPLOYMENT AND DESCENT

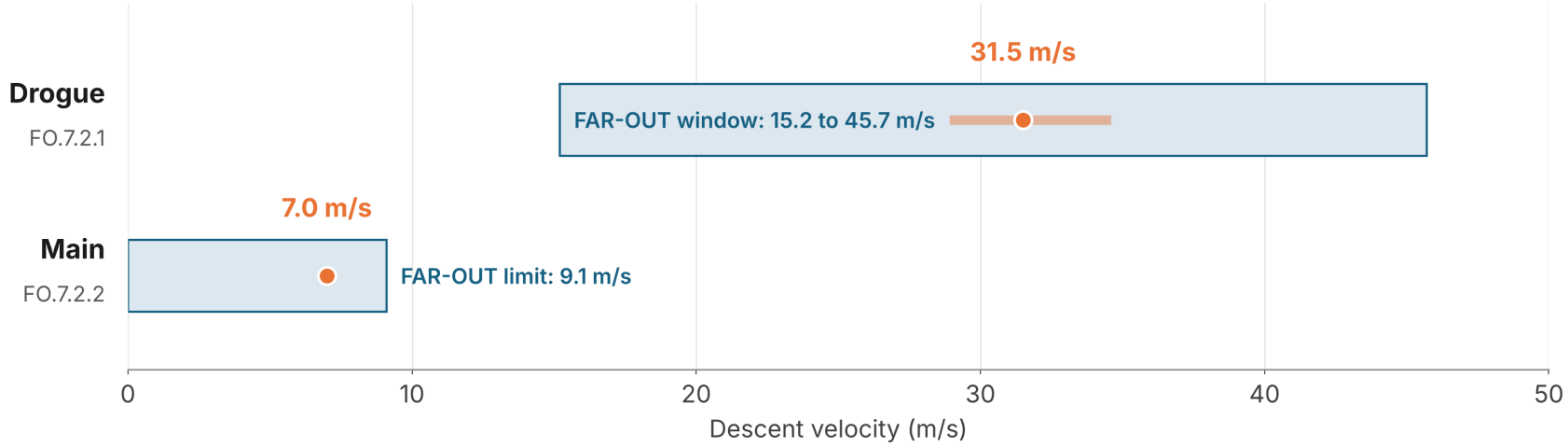
Code	Requirement	Our design	Status
FO.7.1	Two-phase recovery: one drogue, one main	Cross drogue, annular main	Met
FO.7.2.1	Drogue at apogee, 15.2 to 45.7 m/s (50 to 150 ft/s)	31.5 m/s under the drogue	Met
FO.7.2.2	Slow to ≤ 9.1 m/s (30 ft/s) before 305 m (1,000 ft)	Main at ~450 m, 7.0m/s	Met
FO.7.2.3	Above $\frac{1}{3}$ of target apogee, stay above 15.2 m/s	Drogue only up high, 31.5 m/s	Met
FO.1.2.2	Nylon shear pins hold a 15-psi difference	2 x M3 nylon pins per joint into brass M3 nuts, hold 19.8 psi	Met
FO.1.2.3	Charge to ≥ 20 psi, force ≥ 3x total pin strength	65 psi charges, 3.3 x pin strength (1,938 N)	Met

Recovery – Requirements

FAR-OUT MANDATORY: RECOVERY HARDWARE

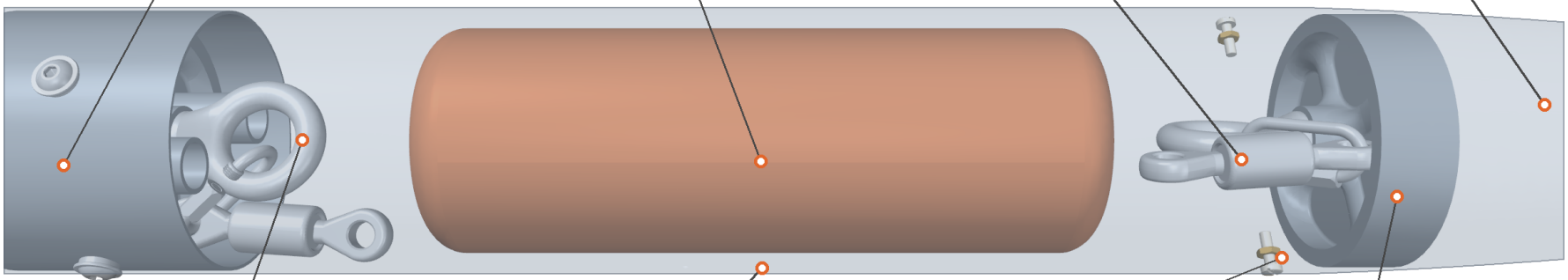
Code	Requirement	Our design	Status
FO.7.3.1	Closed eye bolts, washers, removable thread locker	M10 316 SS eye bolts	Met: thread locker at build
FO.7.3.2	Load-spreading plate on the bulkhead	Eye bolts thread into aluminum bulkheads	Met: bulkhead is the plate
FO.7.3.3	Barrel quick links, safety wired; no carabiners	Oval screw-lock quick links	Met: safety wire at build
FO.7.3.4	Swivel on each parachute, shock cord, or harness	6 swivels: each eye bolt and chute	Met
FO.7.3.5	Non-elastic cords, rated above the hardware	Tubular Kevlar; 8,450 N	Met: above every link
FO.7.10	FOS ≥ 2 on the drogue drag force	Late drogue opening 186 N (394 N design)	Met: FOS 19.1 (9.0 on 394)
FO.1.2.1	Coupler ≥ 1 body caliber into each airframe	110 mm bonded; 81 mm separated	Met: 1.06 caliber on separated end

Recovery – Descent Rates vs FAR-OUT



Both stages pass: the drogue sits mid-window and the main lands 2.1 m/s under the limit

Recovery – Main Bay



Forward recovery bulkhead
Aluminum, in a 60 mm sleeve; charge wells

Main parachute, packed
Annular, D0 1.65 m

Swivel + quick link
8928T41 swivel, 8947T14 link

Nose cone
Von Kármán, G12 fiberglass

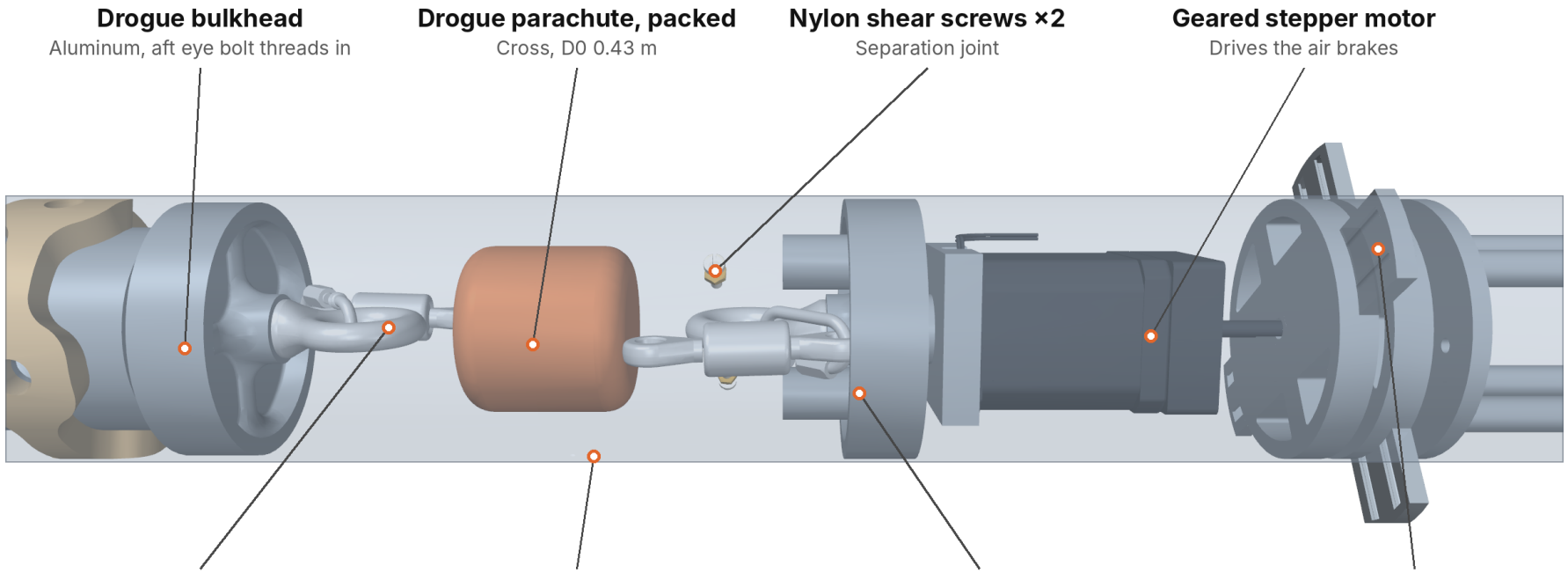
Eye bolt, swivel, quick link
M10 316 SS eye bolt

Upper airframe
G12, 3 in; Ø2 mm vent

Nylon shear screws ×2
Hold the nose cone on

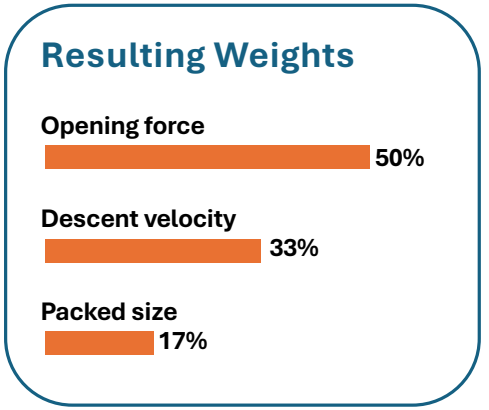
Nose cone bulkhead
Aluminum, forward eye bolt threads in

Recovery – Drogue Bay



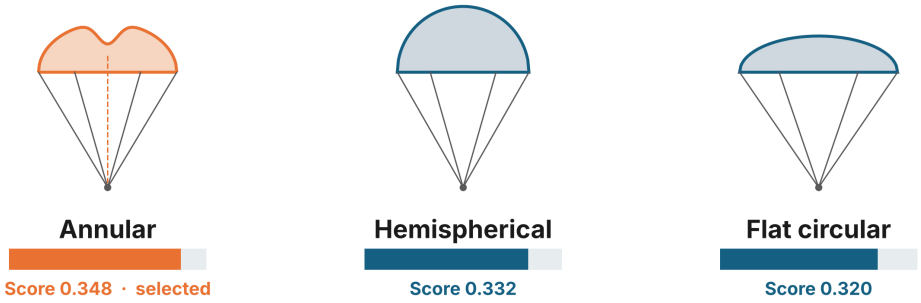
Recovery – Trade Study Weights (BDM)

Comparison	More important	Why
Opening force vs descent velocity	Opening force	Sets the load on every link
Opening force vs packed size	Opening force	Sets the load on every link
Opening force vs mass	Opening force	Sets the load on every link
Descent velocity vs packed size	Descent velocity	FAR-OUT descent limits
Descent velocity vs mass	Descent velocity	FAR-OUT descent limits
Packed size vs mass	Packed size	Chute must fit the bay



Weight = comparisons won ÷ 6. Both parachute trades use these weights; mass loses every comparison, so it carries 0%.

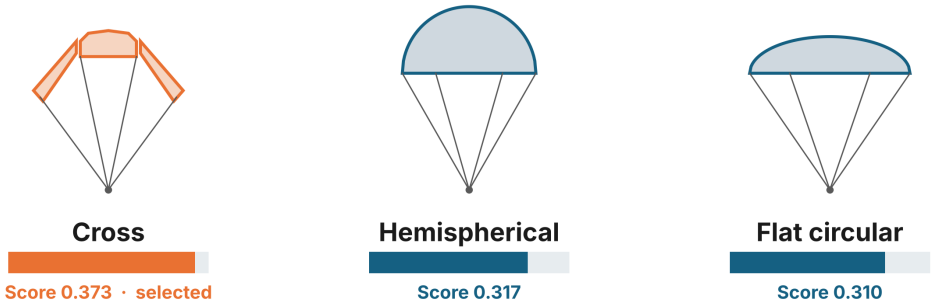
Recovery – Main Parachute Trade Study



Criterion	Weight	Annular (selected)	Hemispherical	Flat circular
Opening force (N)	50%	507 (score: 0.177)	532 (score: 0.169)	585 (score: 0.154)
Descent velocity (m/s)	33.3%	7.0 (score: 0.112)	7.1 (score: 0.110)	7.0 (score: 0.112)
Packed size, D0 (m)	16.7%	1.65 (score: 0.059)	1.85 (score: 0.053)	1.78 (score: 0.055)
Mass (g)	0%	176	179	167
Overall score	**100%**	**0.348**	**0.332**	**0.320**

Values at the final 5.85 kg. Scores are ratios, so the ranking matches the original 4.9 kg run.

Recovery – Drogue Parachute Trade Study



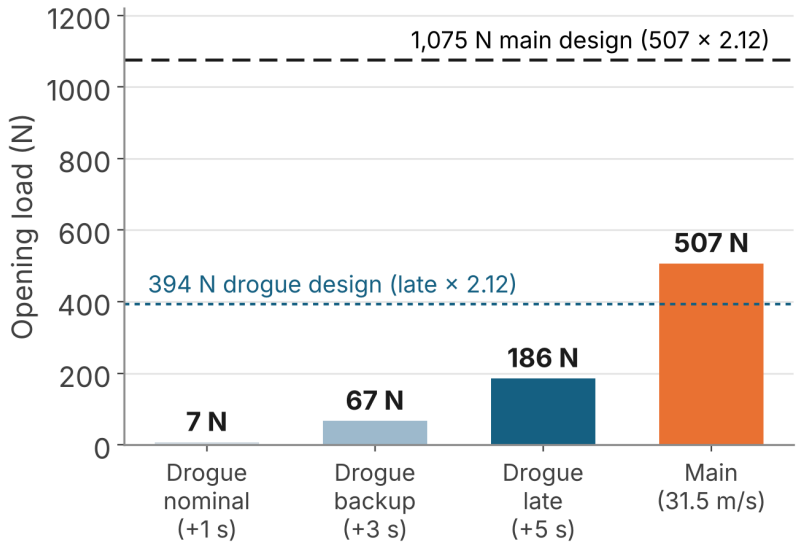
Criterion	Weight	Cross (selected)	Hemispherical	Flat circular
Opening force (N)	50%	7.0 (score: 0.206)	9.5 (score: 0.152)	10.1 (score: 0.142)
Descent velocity (m/s)	33.3%	31.5 (score: 0.112)	32.2 (score: 0.110)	32.0 (score: 0.111)
Packed size, D0 (m)	16.7%	0.43 (score: 0.055)	0.43 (score: 0.055)	0.41 (score: 0.057)
Mass (g)	0%	20	20	19
Overall score	**100%**	**0.373**	**0.317**	**0.310**

Values at the final 5.85 kg. Scores are ratios, so the ranking matches the original 4.9 kg run.

Recovery – Drogue Opening Load

Input	Value	Basis
CdS	0.105 m^2	Cd 0.725 × S0, trade study
D0	0.43 m	Cross canopy
Cx	1.2	Cross, Knacke Table 5-1
n	11.7	Knacke Table 5-6
ρ	1.225 kg/m^3	Sea level, conservative
A = 2m / (CdS·ρ·n·D0)	18.08	A ≫ 1, so X1 ~ 1.0

Case	Deploys	v	q = ½ρv^2	F
Nominal, primary charge	apogee + 1 s	9.8 m/s	59 Pa	7 N
Backup charge	+ 3 s	29.4 m/s	530 Pa	67 N
Late, worst case	**+ 5 s**	**49.1 m/s**	**1,474 Pa**	**186 N**
Drogue design load	**late × 2.12**			**394 N**

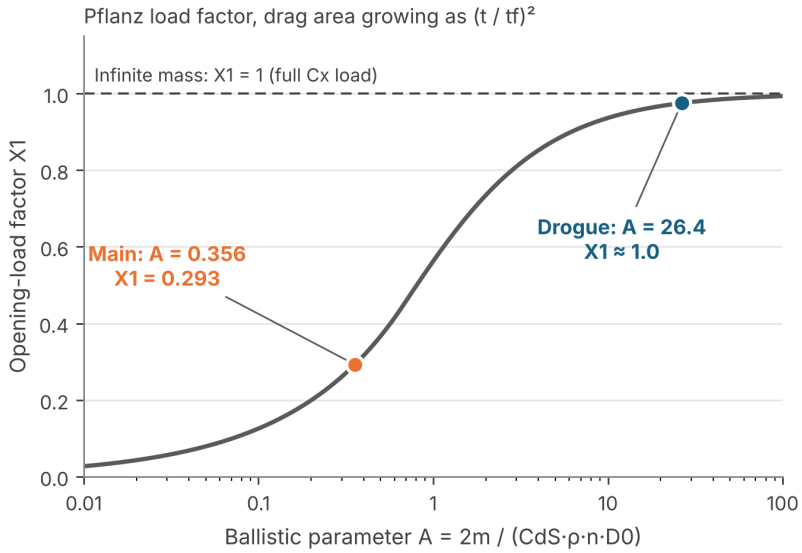


Late deployment governs the drogue: 186 N × 2.12 = 394, the drogue-chain load used in the requirement and link checks.

Recovery – Main Opening Load

Input	Value	Basis
CdS (load case)	2.031 m^2	Cd 0.95 × S0 2.138 m^2, annular
D0	1.65 m	Annular canopy
Cx	1.4	Annular, Knacke Table 5-1
n	8	Knacke Table 5-6
ρ	1.225 kg/m^3	Sea level, conservative
A = 2m / (CdS·ρ·n·D0)	0.356	m = 5.85 kg; A < 1, so X1 = 0.293

Step	Equation	Result
Deploy speed	Drogue descent rate at 450 m	31.5 m/s
$q = ½ρv^2$	½ × 1.225 × 31.5^2	608 Pa
Fill time tf = n·D0 / v	8 × 1.65 / 31.5	0.42 s
Full-Cx load	CdS·q·Cx = 2.031 × 608 × 1.4	1,729 N
Opening load F	**1,729 × X1 = 1,729 × 0.293**	**507 N**
Main design load	**507 × 2.12**	**1,075 N**

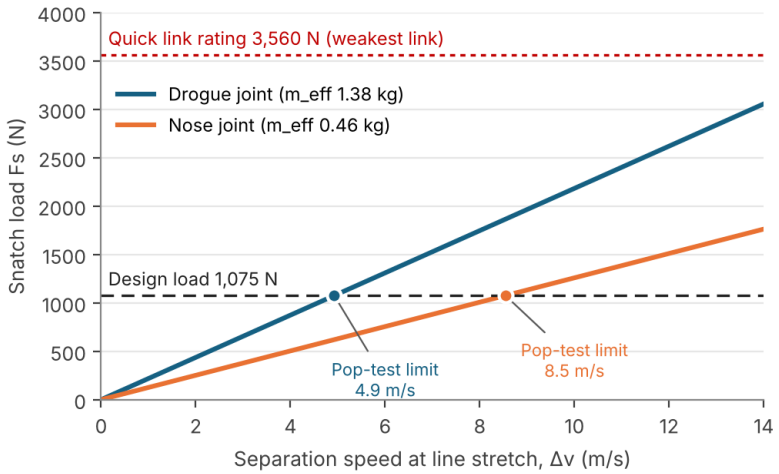


Low mass governs the main: with A = 0.356 the canopy slows the rocket while it fills, so X1 = 0.293 cuts the 1,729 N full-Cx load to 507 N. Multiplying by 2.12 gives the 1,075 N main-chain design load.

Recovery – Snatch Loads

Shock cord	Value	Basis
Break strength Fb	8,450 N	LOC KV-25-30, ¼ in Kevlar
Elongation at break ε	4%	Assumed; Kevlar braid 3–6%
Length L	6.1 m	20 ft each
Stiffness k = Fb / (ε·L)	34.6 kN/m	

Joint	m1 / m2	m_eff	Δv for 1,075 N
Drogue	2.21 / 3.64 kg	1.38 kg	4.9 m/s
Nose	0.50 / 5.35 kg	0.46 kg	8.5 m/s



Snatch can govern. With no losses the charge could separate the sections faster than the pop-test limits. The 3,560 N quick link is the weakest link. The pop test measures Δv; if it is over the limit, cut powder or reconsider design choices.

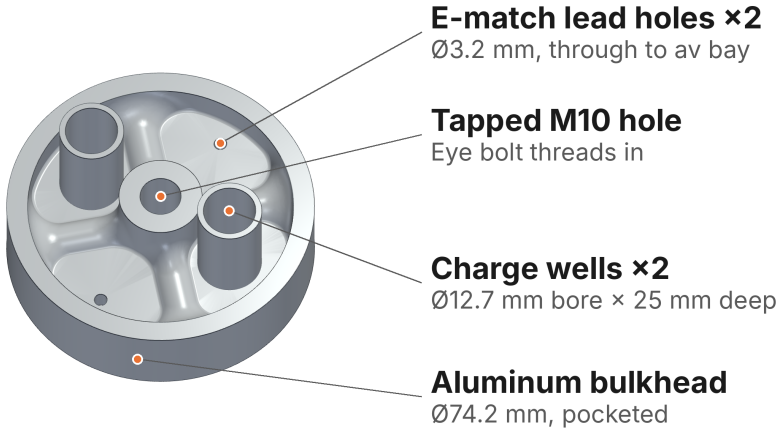
Recovery – Link Load Check

Part	Quantity	Rated load	Factor of Safety, main chain (1,075 N)	Factor of Safety, drogue chain (394 N)
Shock cord (Kevlar)	2	8,450 N	**7.9**	**21.4**
Eye bolt	4	5,338 N	**5.0**	**13.5**
Quick link	6	4003.4 N	**3.72**	**10.16**
Swivel	6	4,893 N	**4.6**	**12.4**

Weakest link: the quick links, FOS of 3.3 on the 1,075 N design load.

Recovery – Charge Wells and BP Sizing

	Formula	Drogue bay	Main bay
Bay volume	$V_B = \frac{\pi}{4} D^2 L$	680 cm^3	1,351 cm^3
Free volume	$V_F = V_B - V_{chute} - V_{hardware}$	458 cm^3	626 cm^3
Target pressure	$P_{target} \geq \frac{3(590\ N)}{29.81\ N/psi}$	65 psi	65 psi
Primary charge	$m_p = \frac{PV}{RT}$	0.94 g	1.28 g
Backup charge	$m_b = 1.25 m_p$	1.17 g	1.60 g
Ejection force	$F_E = P_{target}(6.70\ in^2)$	1,938 N	1,938 N

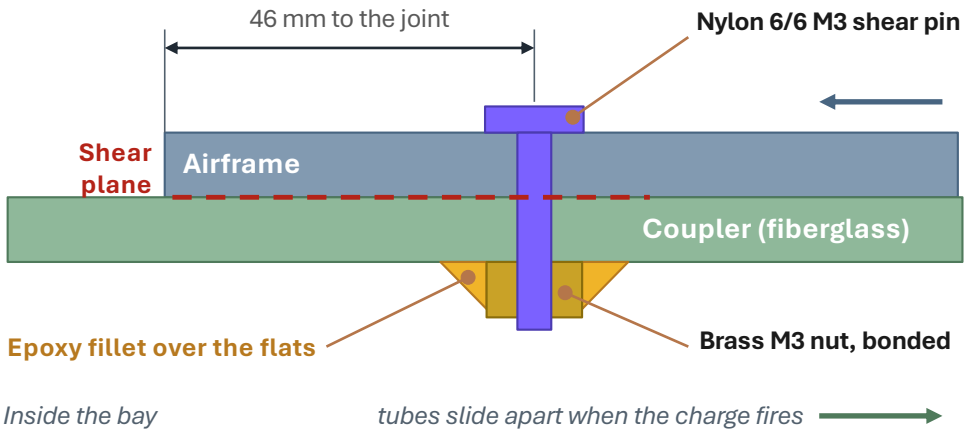


The 1.60 g backup fills about half a well.

Sized to 65 psi: 1,938 N on the ⌀74.2 bore (6.70 in^2) is 3.3x the 590 N pin strength, as FO.1.2.3 requires

Recovery – Shear Pins and Attachment

Section through one pin (not to scale)



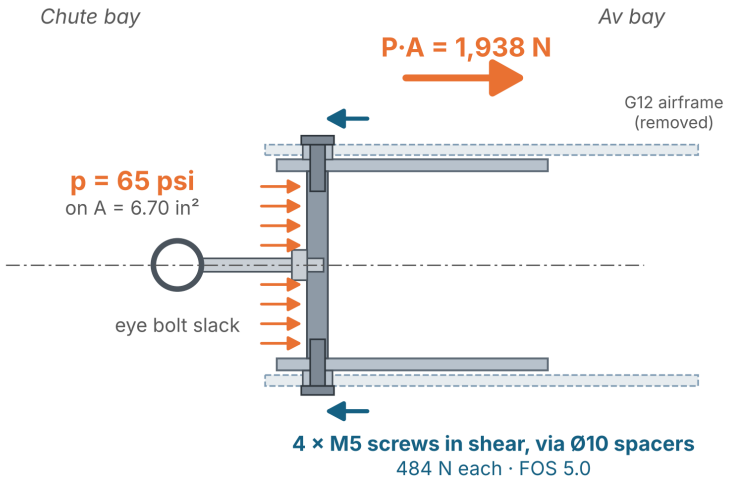
The nut only keeps the pin in. The shear load goes from the pin into the 3.2 mm close-fit holes in both walls.

Check	Result	Need	FS / MS
Pin shear, 4.48 mm² root × 66 MPa	295 N each, 590 N/joint	–	–
Pressure the pins hold	19.8 psi	≥ 15 psi	MS 0.32
Charge force ÷ pin strength	1,938 ÷ 590 = 3.3	≥ 3	Met
Wall bearing when pin breaks	98 MPa	241 MPa	FS 2.4
Wall shear-out, 35 mm edge	4.2 MPa	148 MPa	FS 35
Nut fillet vs install torque	0.39 N·m	0.10 N·m	FS 3.9

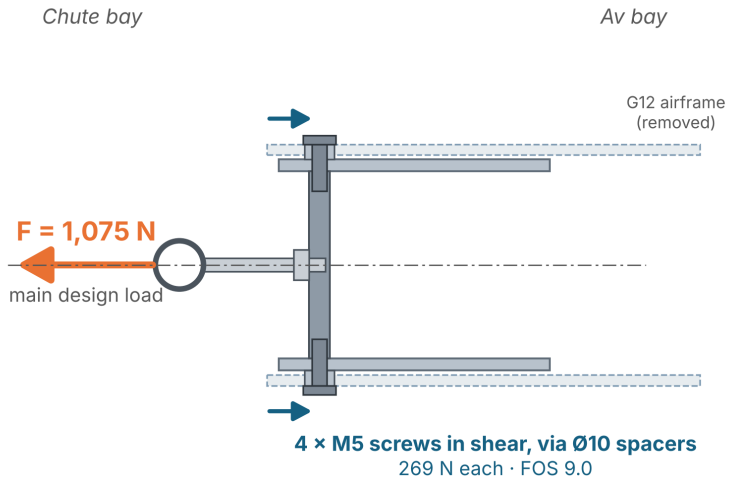
How the nut is held: an ES6209 fillet bonds the brass nut to the inside of the coupler (and the nose shoulder), so it cannot spin or drop out when the pin is threaded in from outside.
Backup: heat-set threaded insert in a G10 doubler, as FO.1.2.2 allows.

Recovery – Bulkhead Load Cases

(a) Ejection charge fires



(b) Main chute opens



Ejection load path

Charge: 65 psi
on the Ø74.2 bore, 6.70 in²

→

Bulkhead: 1,938 N
web 17.1 MPa, FOS 14.1

→

Into the airframe
aft: rim bond FOS 2.9 hot; fwd: 4 × M5 + Ø10 spacers, FOS 5.0

→

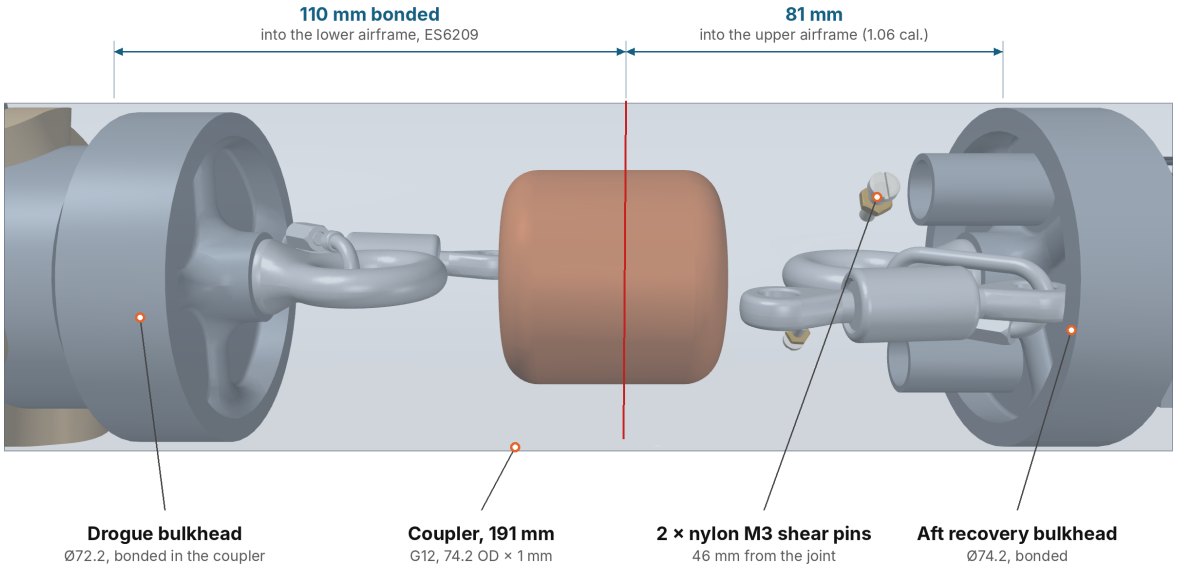
Shear pins: 590 N
break at 19.8 psi; charge is 3.3× pins

Recovery – Structural Checks

Connection	Load case	Stress	Allowable	FOS
Bulkhead web, plate bending	65 psi	17.1 MPa	241 MPa	14.1
Bulkhead hub under the eye bolt	1,075 N	39 MPa	241 MPa	6.2
M10 thread strip in the bulkhead	1,075 N	4.6 MPa	139 MPa	30.2
Rim bond, aft Ø74.2 bulkhead (ES6209, 130°F)	**1,938 N ejection**	**0.42 MPa**	**1.2 MPa**	**2.9**
Rim bond, Ø72.2 drogue / nose (130°F)	1,835 N ejection	0.40 MPa	1.2 MPa	3.0
Coupler bond, 110 mm (130°F)	1,938 N ejection	0.08 MPa	1.2 MPa	15.9
Fwd bulkhead, 4 × M5 screws: shear	484 N each	38.2 MPa	309 MPa	8.1
Ø10 spacers: bearing on 1 mm G12	484 N each	48.4 MPa	241 MPa	5.0
M5 thread in the bulkhead, 6 mm engaged	484 N each	10.3 MPa	139 MPa	13.5
Av bay thumb screw, Ø10 (sled 0.4 kg at 86 g)	337 N	5.8 MPa	205 MPa	35

Weakest mount: the aft bulkhead's rim bond, FOS 2.9 at 130°F under ejection (MOS 1.9).

Recovery – Couplers and Mounting



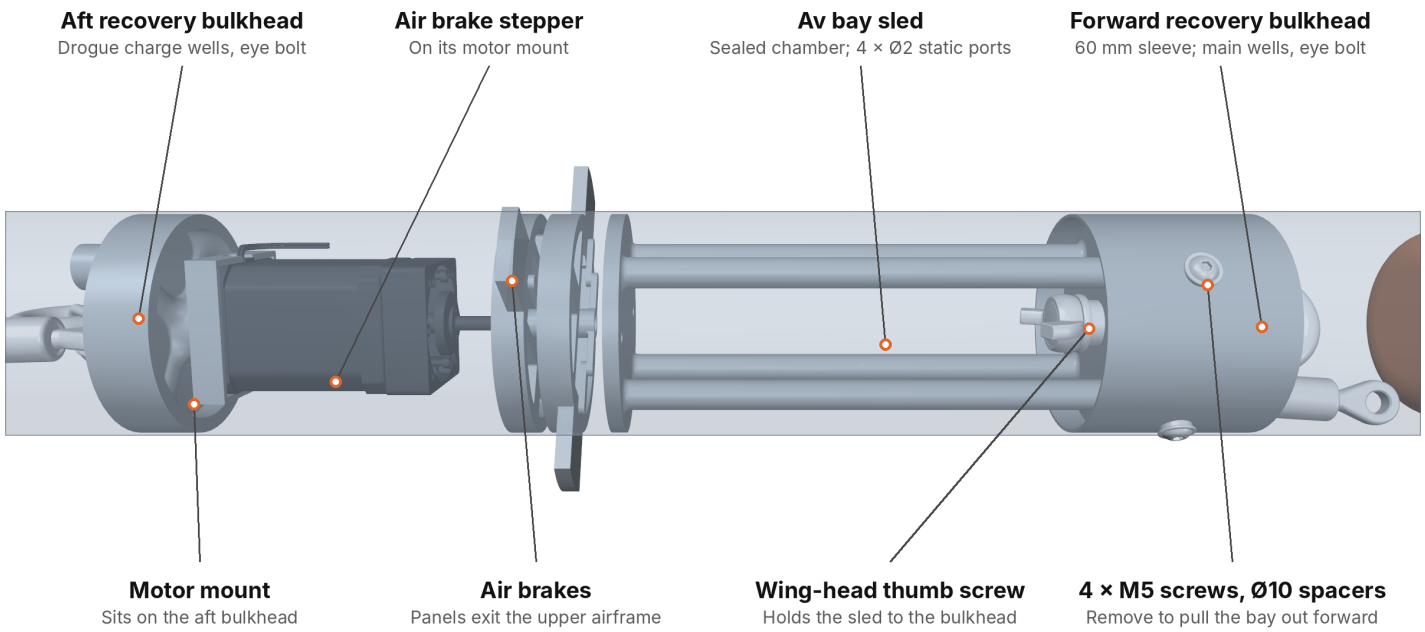
Two joints come apart

Drogue joint at apogee: the coupler slides 81 mm into the upper airframe and is held by 2 x M3 nylon shear pins

Main joint, at 450 m: the nose cone shoulder, held by 2 x M3 nylon shear pins

Screwed, not bonded: the forward recovery (av bay) bulkhead is held by 4 x M5 screws, so it and the av bay pull out forward for arming

Recovery – Integrated Avionics Bay

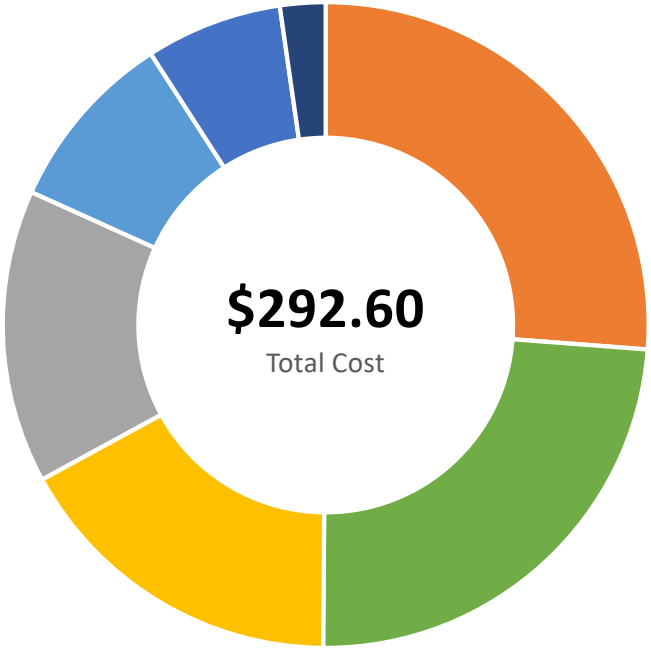


To arm or service: with the nose cone and main out, remove the 4 M5 screws and pull the forward bulkhead and av bay out together. One thumb screw holds the bay to the bulkhead.

Recovery – Materials

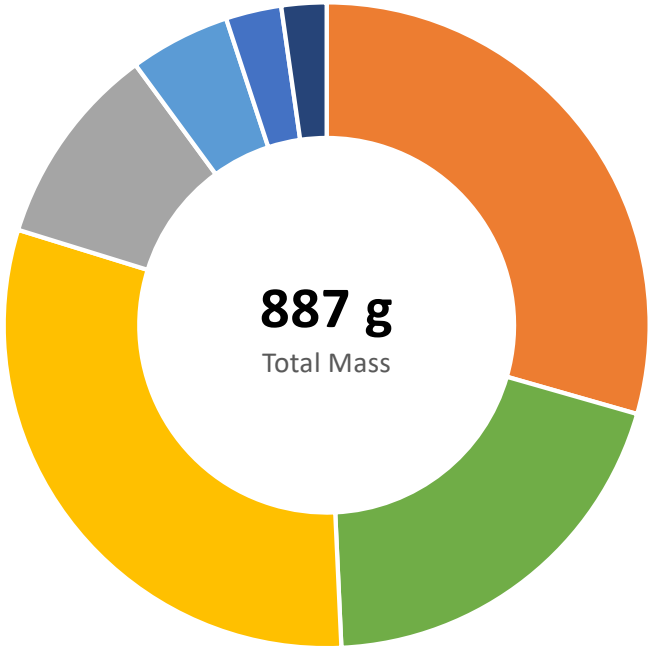
Part	Material	Rating used	Why?
Main parachute	Ripstop nylon, Rocketman Ultra-Light Annular	D0 1.65 m, C_D ≈ 0.9; 507 N opening (1,075 N design)	Trade study winner: lowest opening force, 7.0 m/s descent
Drogue parachute	Ripstop nylon, team-sewn cross	D0 0.43 m, C_D ≈ 0.73; 31.5 m/s descent	Stable, packs small, cheap
Shock cords	¼ in tubular Kevlar, 2 × 6.1 m	8,450 N (1,900 lb)	Non-elastic, heat resistant near the charges
Eye bolts	316 SS, M10 shoulder (McMaster 3032T16)	5,338 N (1,200 lb)	Closed eye, corrosion resistant, FS 5.0
Swivels	316 SS ball bearing eye-eye (McMaster 8928T41)	4,893 N (1,100 lb)	Stops line twist; FS 4.6
Quick links	3/16 in stainless barrel, safety wired (McMaster 8947T25)	4003.4 N (900 lb)	No swing gate, easy rigging, FS 3.3
Bulkheads	6061-T6 aluminum alloy	241 MPa yield; web FS 14.1, bond FS 2.9 at 130°F	Strong, heat resistant, eye bolts thread straight in
Ejection charges	4F black powder, 2 per bay	65 psi: main 1.28 / 1.60 g, drogue 0.94 / 1.17 g	Proven, sized by ideal gas
Shear pins	Nylon 6/6 M3 x 10 mm (McMaster 94270A120), brass M3 nuts	295 N each, 590 N per joint	Breaks clean at ejection, meets requirements

Recovery – Cost Budget



	Item	Source	Cost	%
	Swivels ×6	McMaster 8928T41	$76.68	26%
	Main parachute, 5 ft annular	Rocketman	$70.00	24%
	Shock cords ×2, 20 ft Kevlar	LOC KV-25-30	$49.54	17%
	Powder, e-matches ×4, charge wells ×4	Launch club, MJG Tech	$43.00	15%
	Eye bolts ×4, M10	McMaster 3032T16	$26.72	9%
	Drogue fabric, team-sewn	Ripstop nylon	$20.00	7%
	Quick links ×6	McMaster 8947T14	$6.66	2%
	Total		**$292.60**	100%

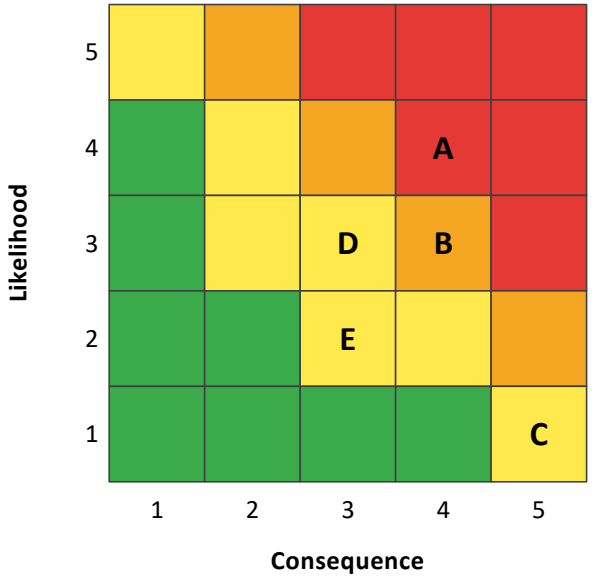
Recovery – Mass Budget



	Component	Basis	Grams	%
	Eye bolts ×4	CAD, 316 SS 8.0 g/cm³	261.1	29%
	Main parachute	Rocketman listing (BOM)	176.0	20%
	Swivels ×6	CAD, 316 SS 8.0 g/cm³	270.6	31%
	Shock cords ×2	BOM, 40–50 g each	90.0	10%
	Quick links ×6	CAD, steel 7.85 g/cm³	44.7	5%
	Charges ×4 and e-matches ×4	Sized for 65 psi; BOM e-matches	24.7	3%
	Drogue parachute	Chute trade study	20.0	2%
	Total		**887.1**	100%

Recovery – Risk Assessment

ID	Failure mode → effect	L×C	Score	Action → after
A	Charge or e-match fails → chute stays packed, hard landing	4×4	**16** Critical	Backup charges + ground test → **8**
B	Shear pins break early → bay opens at high speed	3×4	**12** High	Pin strength calc + ground test → **8**
C	Quick link breaks at main opening (FOS 3.3) → main lost	1×5	**5** Medium	Safety wire; 1,075 N load test. Accept
D	Powder burns canopy or cord → torn chute, faster landing	3×3	**9** Medium	Nomex blanket, careful packing → **6**
E	Lines twist or tangle → chute only partly opens	2×3	**6** Medium	Swivels at every link; packing practice. Accept



Recovery – Next Steps

Estimating six weeks from PDR to CDR

Week 1
- PDR Feedback
- Order parts ($293)

Week 2
- Prep ground tests

Week 3
- Sew cross drogue
- Parts arrive

Week 4
- Pop and load tests
- Nomex blanket fit

Week 5
- Vacuum tests
- Update all design choices

Week 6
- Create CDR slides
- Presentation rehearsals

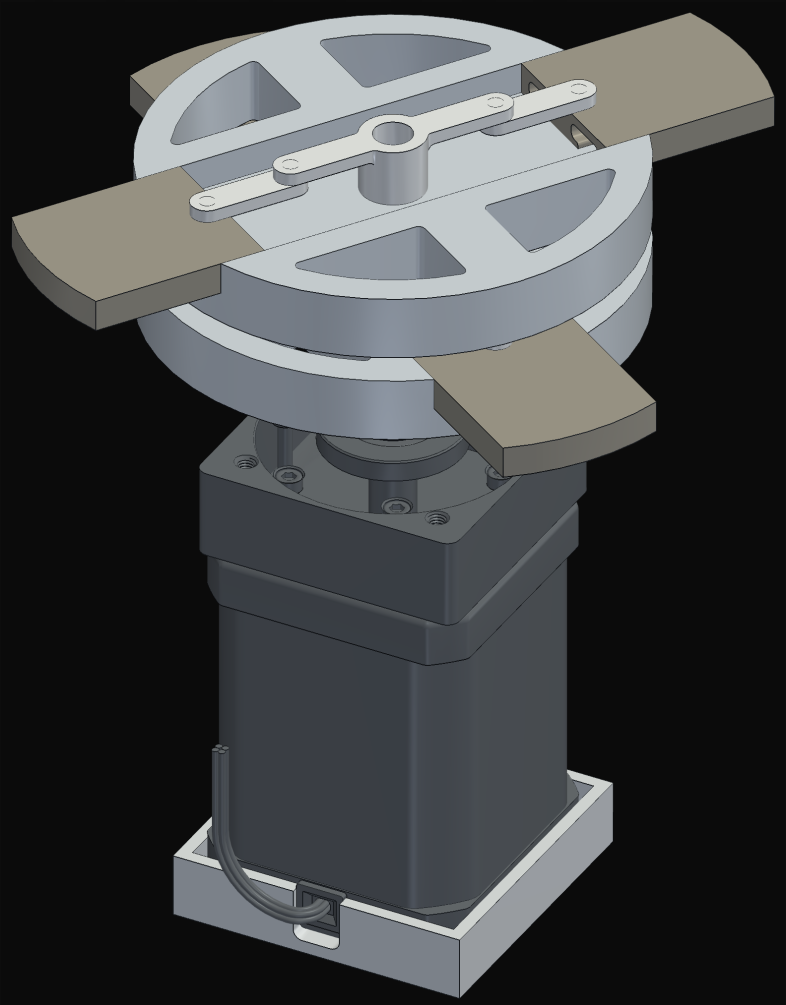
Pop test
Pass if:
- Pins shear, chute fully out
- Separation ≤ 4.9 m/s (drogue), ≤ 8.5 m/s (nose)
- No damage, 2 passes per bay

Load test
Pass if:
- Pull bulkhead to 1,075 N
- Hold 10 s
- No cracking or slip

Vacuum test
Pass if:
- Drogue fires at apogee
- Main fires at set altitude
- Both altimeters

Any fail: adjust and retest. No flight until all three pass. CDR at the end of week 6.

Airbrakes



Airbrakes – Requirements

Code	Requirement	Our design	Status
FO.2.7	Active aerosystems pre-programmed, keep a vertical trajectory	All 4 panels run off one crank, so drag stays symmetric	Partly met: control code for CDR
FO.4.3	Attachments removable and not critical structure, unless an air-brake	Track plates bonded into the airframe, allowed for air-brake	Met
FO.4.4.1	Airbrakes count toward payload weight	Motor and gearbox 1.41 kg, plus plates, panels, and linkage	Met: tracked in payload mass
FO.2.3.2	Contract apogee set at the first Progress Update	Panels trim apogee down to the contract altitude	Open: control law in sim
FO.6.9.1	Vehicle schematics include active control wiring	Stepper and driver wiring goes in the vehicle schematic	Open: due 3rd Progress Update

Airbrakes – Design

Links ×4
13.7 mm, pin to pin

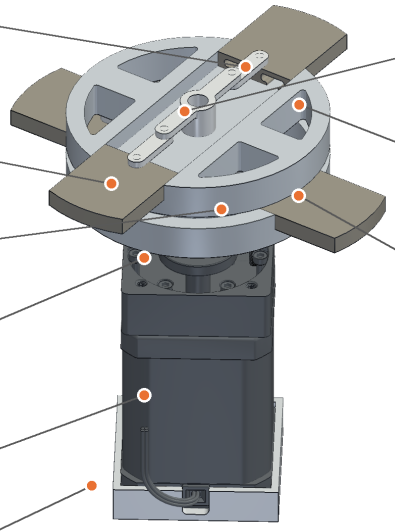
Brake panels ×4
Steel, 25 × 30 × 5 mm

Epoxy bond
ES6209 to airframe, 41.5 cm²

5:1 planetary gearbox
0.36° per step at output

NEMA 17 stepper
3.95 N·m, 1.41 kg with gearbox

Motor mount
Holds the motor on axis

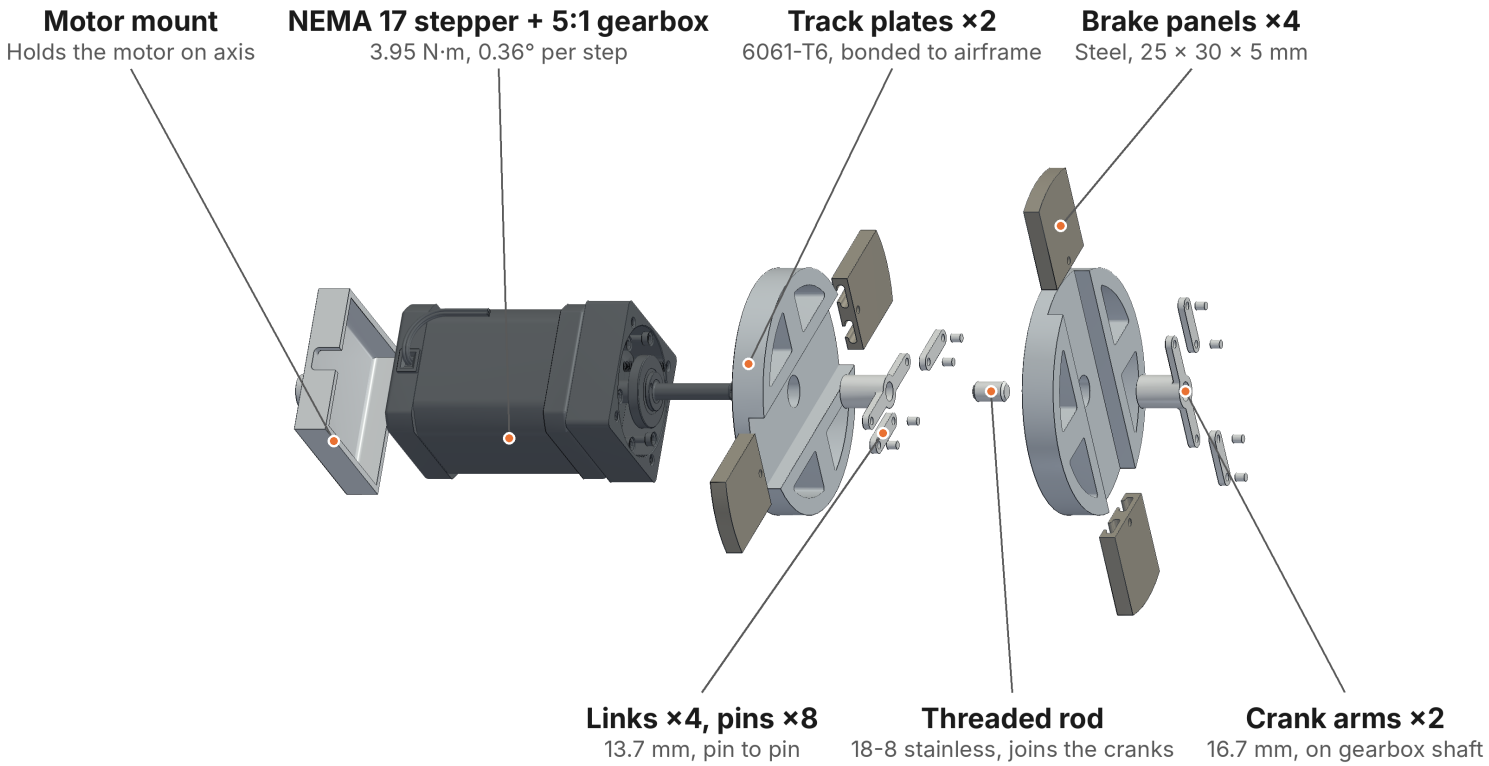


Crank arms ×2
16.7 mm, on gearbox shaft

Track plates ×2
6061-T6

Dry PTFE lube
Panels, track walls, slot edges

Airbrakes – Exploded View



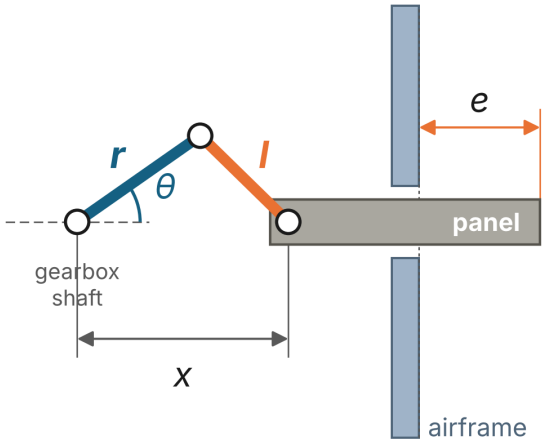
Airbrakes – Inverse Kinematics

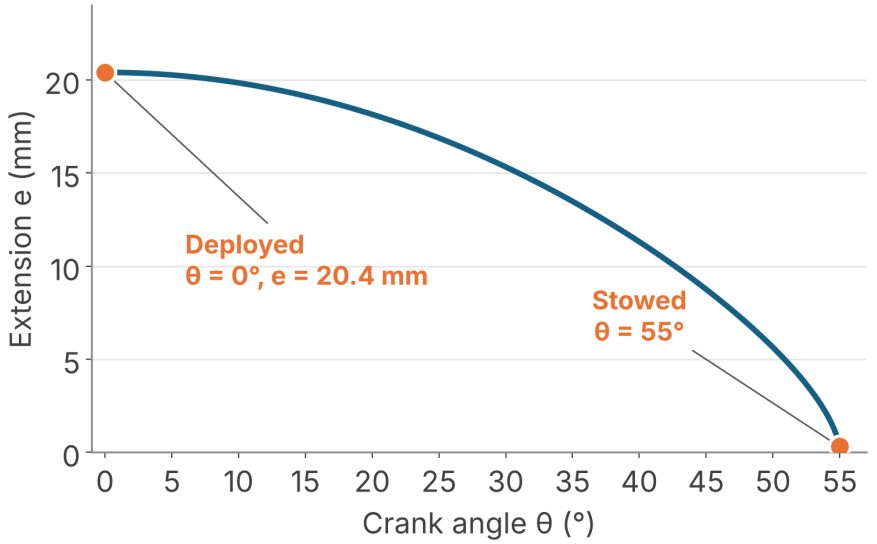
$$x = r\cos\theta + \sqrt{l^2 - r^2\sin^2\theta}$$

$$\theta = \arccos\left(\frac{r^2 + x^2 - l^2}{2rx}\right)$$

$$e = x - x_{\text{stowed}}$$

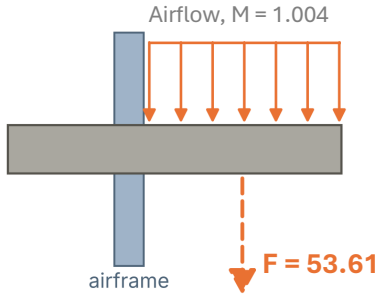
$r = 16.7$ mm, $l = 13.7$ mm



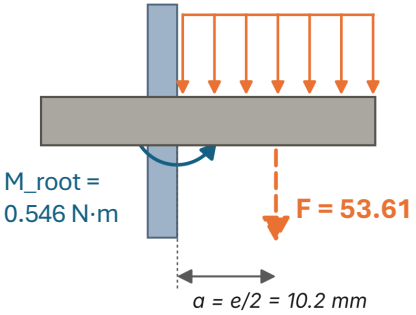


Stowed at 55°, the link sits nearly square to the panel, so drag can't back-drive the crank

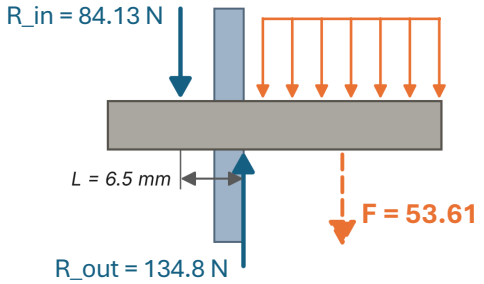
Airbrakes – Panel Loads



Aero drag	
$q = \frac{1}{2}\rho v^2$	64,262Pa
K_c	1.278
$q_c = K_c\, q$	82,129.4 Pa
$A = \mathrm{lw}$	5.1 cm²
$F = q_c\, C_d\, A$	53.61 N
$F_{\mathrm{total}} = 4F$	214.46 N

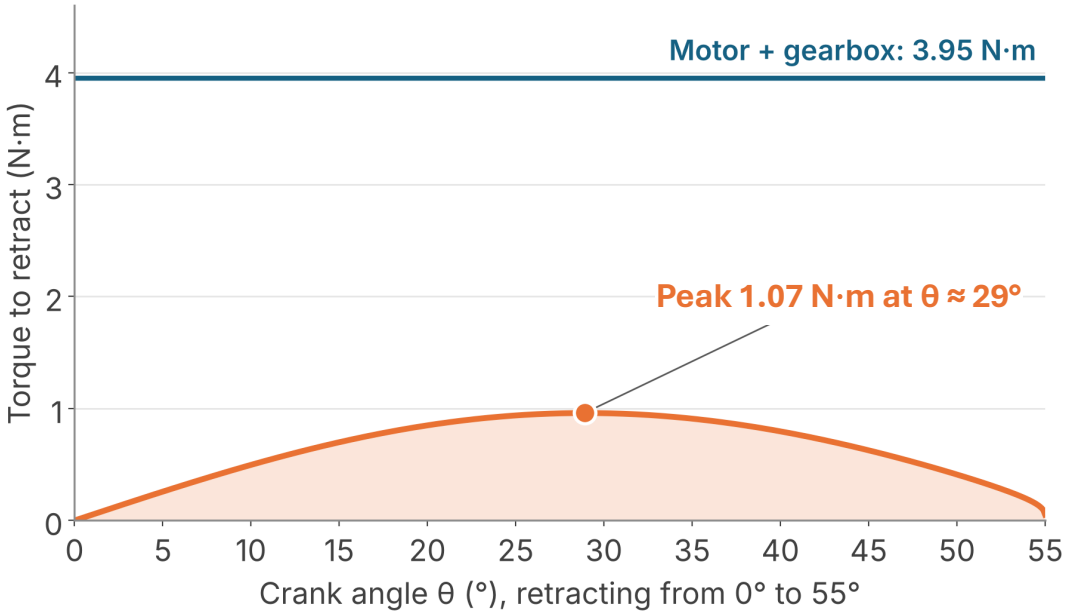


Panel building	
$M = F\,\frac{e}{2}$	0.546 N·m
$I = \frac{bh^3}{12}$	2.60e-10 m⁴
$\sigma = \frac{Mc}{I}$	5.25 MPa
Steel yield	210 MPa
Factor of safety	40



Track friction	
$R_{\mathrm{in}} = \frac{F\,a}{L}$	84.13 N
$R_{\mathrm{out}} = F + R_{\mathrm{in}}$	134.75 N
$F_f = \mu(R_{\mathrm{in}} + R_{\mathrm{out}})$	22.19 N
Bearing stress	9.18 MPa
6061-T6 track yield	241 MPa
Factor of safety	26.2

Airbrakes – Motor Torque



Retract, 4 panels	
Friction per panel, full out	22.19 N
$T(\theta) = 4\,F_f(\theta)\left\|\frac{dx}{d\theta}\right\|$	At each θ
Peak, at θ ≈ 29°	1.07 N·m
Motor + gearbox	3.95 N·m
Factor of safety	3.9

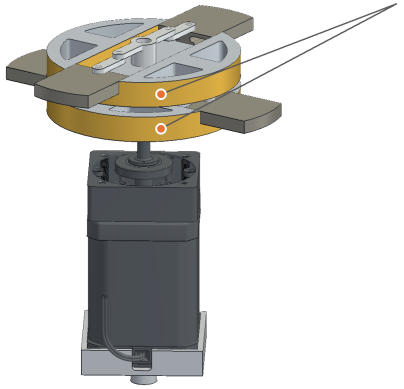
Drag pushes along the rocket axis, square to the slide, so only track friction fights the motor.

Airbrakes – Mounting

Condition	Lap shear	Knockdown	Allowable
77°F	20.0 MPa	25%	5.0 MPa
130°F	4.8 MPa	25%	1.2 MPa

Joint	Load	Area	Shear	FS 77°F	FS 130°F
Plates, ejection	1,938 N	41.5 cm^2	0.47 MPa	10.7	2.6
Plates, opening	1,075 N	41.5 cm^2	0.26 MPa	19.3	4.6

Allowable = 25% of the Aeropoxy lap shear at that temperature



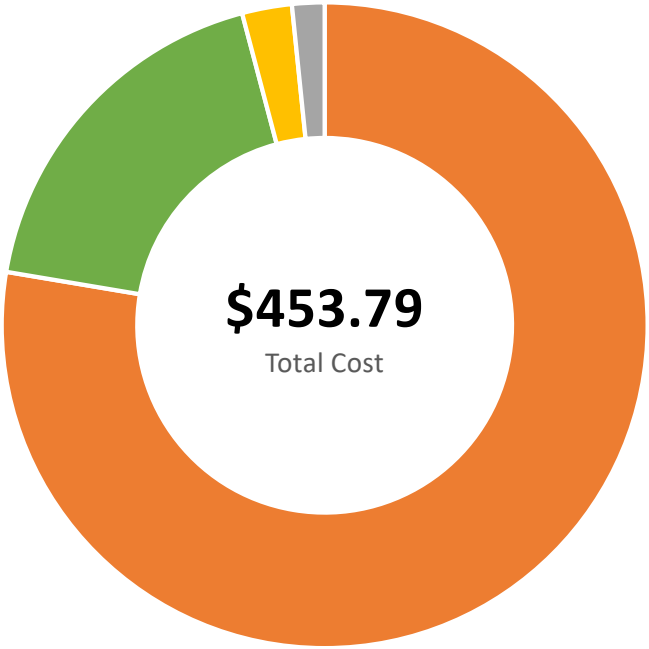
Bonded faces
Both plate rims, 41.5 cm^2
ES6209 to the airframe

Lowest factor of safety is 2.6, at 130°F under the 1,938 N ejection load.

Airbrakes – Materials

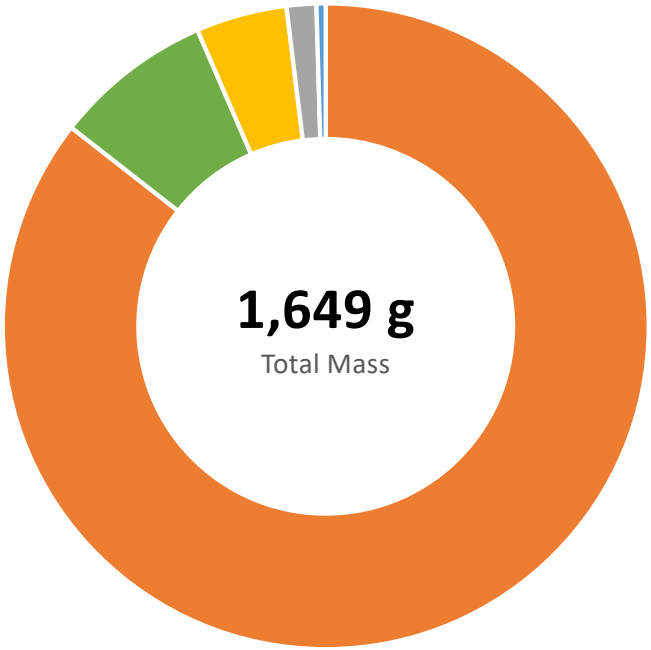
Part	Material	Rating used	Why?
Track plates ×2	6061-T6 aluminum, bonded to the airframe	241 MPa yield; bond FS 2.6 at 130°F	Light, easy to machine the panel slots, bonds well with ES6209
Brake panels ×4	AISI 1020 steel, 25 × 30 × 5 mm, painted	210 MPa yield; root bending 5.13 MPa, FS 41	Hard-wearing sliding face in the aluminum track; cheap stock
Cranks ×2, links ×4	6061-T6 aluminum	241 MPa yield	Light; cut from the same plate stock
Bond	Aeropoxy ES6209	1.2 MPa at 130°F (25% of lap shear)	Same adhesive as the recovery bulkheads
Actuator	Geared NEMA 17 stepper, 5:1 (McMaster 4801N12)	3.95 N·m; FS 3.9 on the 1.01 N·m peak	Open-loop position control with enough torque at Mach 0.995
Lubricant	Dry PTFE spray	–	Dry film; doesn't collect grit in the slots

Airbrakes – Cost Budget



	Item	Source	Cost	%
	Geared stepper, NEMA 17, 5:1	McMaster 4801N12	$352.36	78%
	Aeropoxy ES6209, quart kit	Aircraft Spruce	$82.75	18%
	1020 steel, 30 × 30 × 5 mm stock (4 panels)	McMaster 89015K172	$11.32	2%
	Dry PTFE lube spray, 10 oz	WD-40 Specialist 300059	$7.36	2%
	Total		**$453.79**	100%

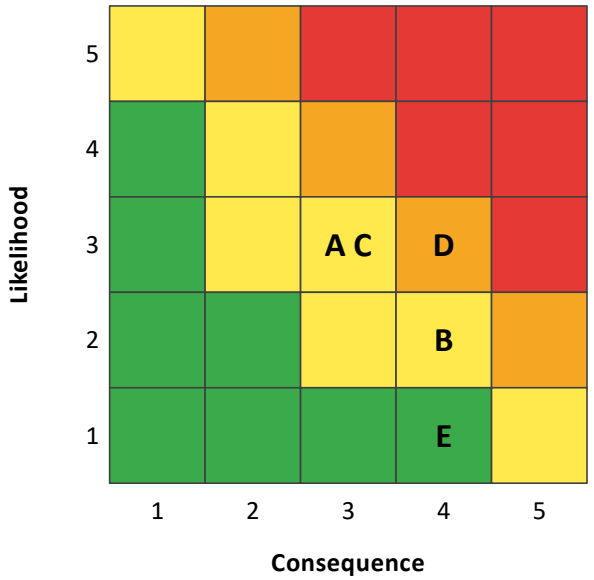
Airbrakes – Mass Budget



	Component	Basis	Grams	%
	Motor + 5:1 gearbox	McMaster 4801N12, 3.11 lb	1,410.7	86%
	Track plates ×2	BOM, 6061-T6	130.9	8%
	Brake panels ×4	BOM, AISI 1020 steel	75.1	5%
	Motor mount, pins, rod	CAD; not in BOM yet	24.5	1%
	Cranks ×2, links ×4, ES6209	BOM, 6061-T6	7.6	<1%
	Total		**1,648.8**	100%

Airbrakes – Risk Assessment

ID	Failure mode → effect	L×C	Score	Action → after
A	Panel jams in its track → can't deploy or retract, apogee off	3×3	**9** Medium	PTFE lube, slot clearance; bench cycle test → **6**
B	Link or pin breaks → one panel loose, uneven drag	2×4	**8** Medium	Pin and link check; bench cycles → **4**
C	Stepper skips steps → wrong extension, apogee error	3×3	**9** Medium	Home to a stowed limit switch; FS 3.9 torque → **6**
D	Code or wiring deploys during boost → drag at max Mach	3×4	**12** High	Burnout lockout; sim and bench tests → **4**
E	Bond lets go on a hot day → mechanism shifts and jams	1×4	**4** Low	FS 2.6 at 130°F. Accept



Airbrakes – Next Steps

Estimating six weeks from PDR to CDR

Week 1
- PDR Feedback
- Order parts ($454)

Week 2
- Control code
- Burnout lockout
- Add homing switch

Week 3
- Machine plates, cranks, links
- Trim plates to size

Week 4
- Bench cycle test
- Bond plates into position

Week 5
- Torque and lockout tests
- Tune control in sim

Week 6
- Create CDR slides
- Presentation rehearsals

Cycle test
Pass if:
- All 4 panels reach 20.4 mm together
- No jams over 100 cycles
- Stowed position repeats

Torque test
Pass if:
- Retracts against 21.7 N per panel
- No skipped steps
- At our real step rate

Lockout test
Pass if:
- No motion on the pad or in boost
- Deploys only after burnout
- Homes to stowed on reboot

Any fail: adjust and retest. No flight until all three pass. CDR at the end of week 6.

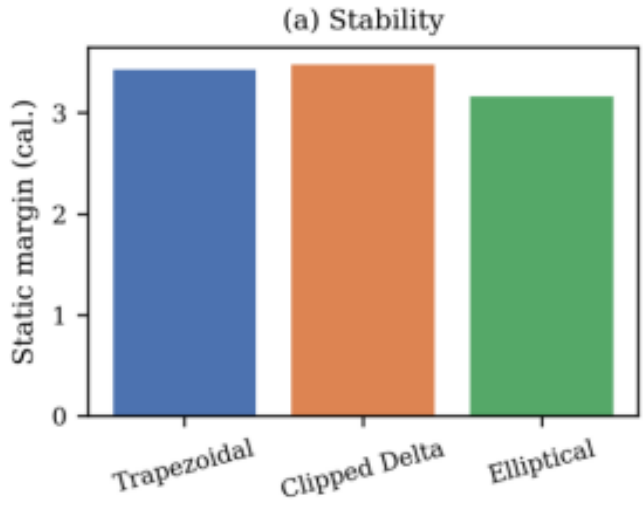
Aerodynamics

Fin Shape Choice

Our fins are Clipped Delta with a root chord of 15 cm and a tip chord of 5 cm with an airfoil shape cross section.

$$M_n = M_\infty \cos \Lambda$$

Clipped Delta Fins reduce shock wave formation by its swept edge (slows the flow down) and have slightly better stability than the trapezoidal fins at the same area.



Fin Shape Justification

- 15 cm root chord:

1.97 Cr/D ratio, which adds strength to the root chord. Capped at Cr/D ratio of 2 to limit skin friction drag from the fins.

- 5 cm tip chord:

Keeps our large leading-edge sweep of ~60 degrees while maintaining maximum lift.

- Airfoil Cross Sectional Shape was chosen to reduce pressure and form drag and increases lift at an Angle of Attack.

Barrowman's Calculator Overview

Used to generate the location of the Center of Pressure by using length of the nose and fin dimensions.

I rearranged these equations in MATLAB to solve for the necessary fin semispan for a stability caliber of 1.5 and achieved a stability caliber of 1.58 in OpenRocket.

Limitations: Higher Mach Ranges, High Angle of attack

Fin Terms

$$(C_N)_F = \left[1 + \frac{R}{S+R}\right]\left[\frac{4N\left(\frac{S}{d}\right)^2}{1+\sqrt{1+\left(\frac{2L_F}{C_R+C_T}\right)^2}}\right]$$

$$X_F = X_B + \frac{X_R}{3}\frac{(C_R+2C_T)}{(C_R+C_T)} + \frac{1}{6}\left[(C_R+C_T) - \frac{(C_R C_T)}{(C_R+C_T)}\right]$$

Finding the Center of Pressure

Sum up coefficients: $(C_N)_R = (C_N)_N + (C_N)_T + (C_N)_F$

Find CP Distance from Nose Tip:

$$\bar{X} = \frac{(C_N)_N X_N + (C_N)_T X_T + (C_N)_F X_F}{(C_N)_R}$$

Flutter Calculations

$$V_f = a\sqrt{\frac{G}{\frac{1.337AR^3P(\lambda+1)}{2(AR+2)(\frac{t}{c})^3}}}$$

Equation 1. Flutter Boundary Equation

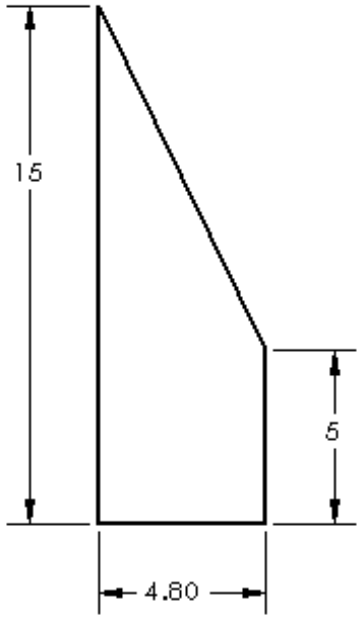
What is Flutter?

I used this equation from an NACA paper that was given to us by our lead to use our max velocity to solve for the thickness our fins need to be to resist flutter

Key components:

- Shear Modulus (G)
- Safety Factor
- Final Calculated of .218 cm thick.

Final Fin Shape with Sized Semispan



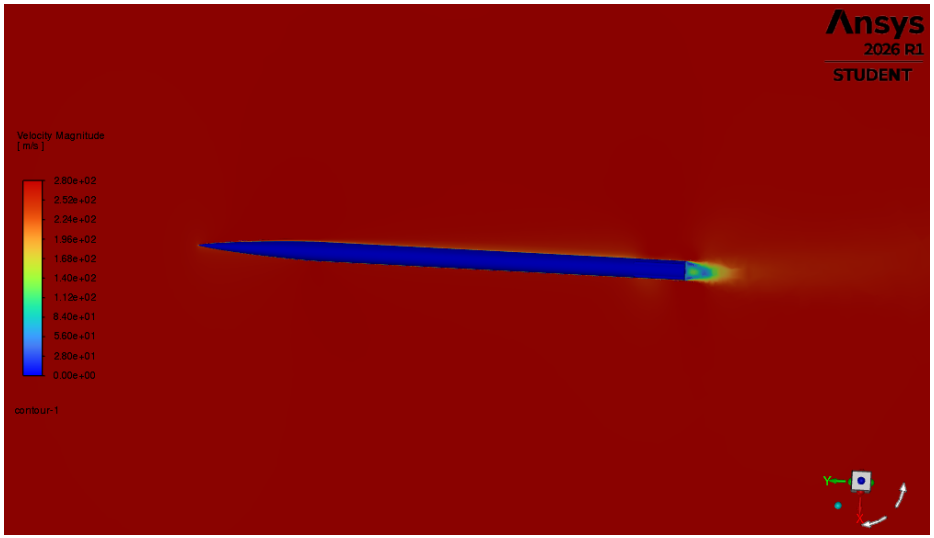
Note on CFD:

Our rocket motor changed on Friday, i.e. CFD results are inaccurate for submitted rocket, but precent error is tailored to the previous rockets design and still accurate for explanations.

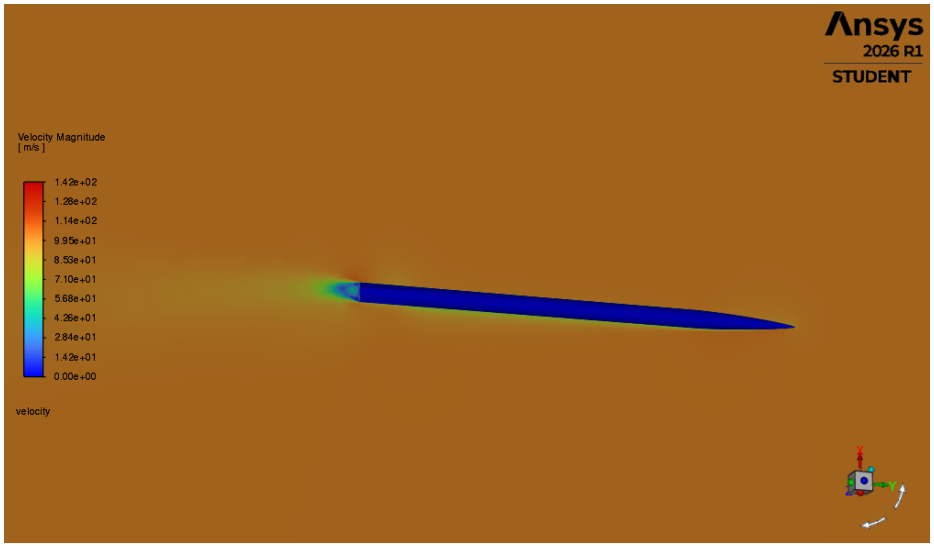
CFD of Fins

Conclusion: Barrowman's Method is inaccurate for higher, transonic region.

Max Speed (.816 Mach):
Cell Count: 73548
Skewness Max: .5471
Orthogonality Min: .200
Wall y+: 48.040251
Cp from nose: 1.496
Percent Error: 10.8%
(estimate: 1.35)



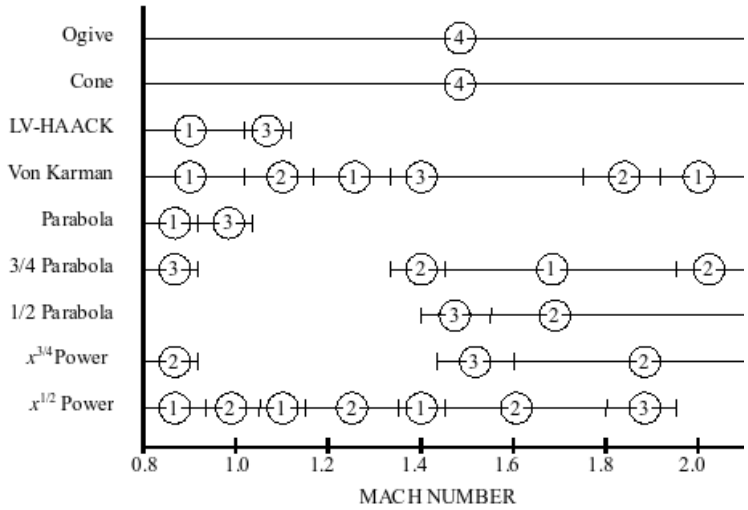
Cell Count: 735721
Cell Skewness Max: 0.54716092
Cell Orthogonality Min: 0.20000363
Wall Y+: 47.986117
Cp From Nose Tip: 1.366
Percent Error: 1.2%
(estimate: 1.35)



Nose Cone Shape Justification

We chose the Von Karman nosecone:

- Pressure drag of 0 (joint angle is 0, low M)
- 16.8% less pressure drag than LV HAACK
- Graph shows superiority in higher machs

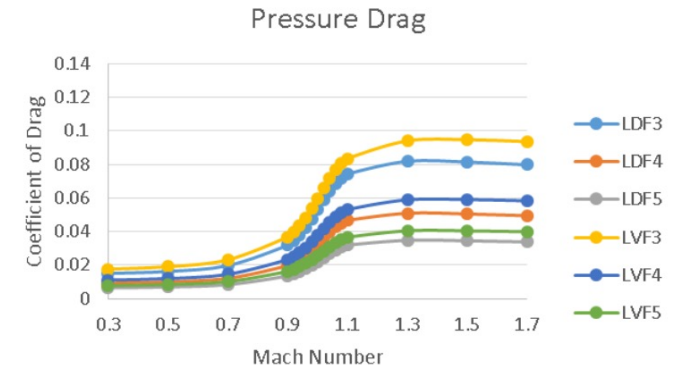


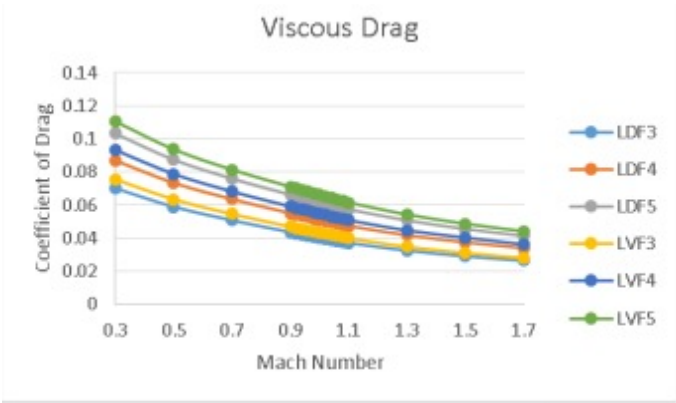
For optimal nonsecone shape for drag, 1 is best.

Nose Cone Fineness Justification

We chose a 4:1 fineness ratio for the nosecone:

- Less Pressure drag than 3:1
- Less Viscous drag than 5:1
- Fineness of 4:1 is a good balance between both
- Also, enough room for any components that are needed to be put inside the nosecone.





Wall Thickness Calculations

Nasa technical reports:

$$\gamma = 1 - 0.901\,(1 - e^{-\phi})$$

Z curvature parameter; $\frac{\ell^2}{rt}\sqrt{1-\mu^2}$ for isotropic cylinder,

$$\phi = \frac{1}{16}\sqrt{\frac{r}{t}} \text{ for } (\frac{r}{t} < 1500)$$

$$k_x = \frac{4\sqrt{3}}{\pi^2}\gamma Z$$

- Original Uncalculated: .2cm
- Calculated: .49 mm
- Used : .1 cm

Nx is the Max Thrust + Max Drag, don't happen at the same time, but it creates a safety factor

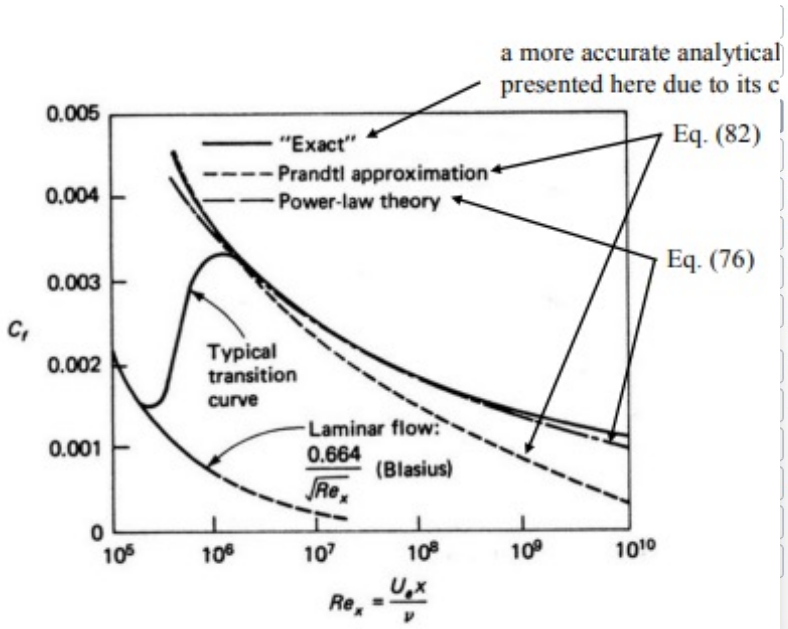
- Limitations: Only accounted for axial compression not bending. Doubled the thickness because of this.

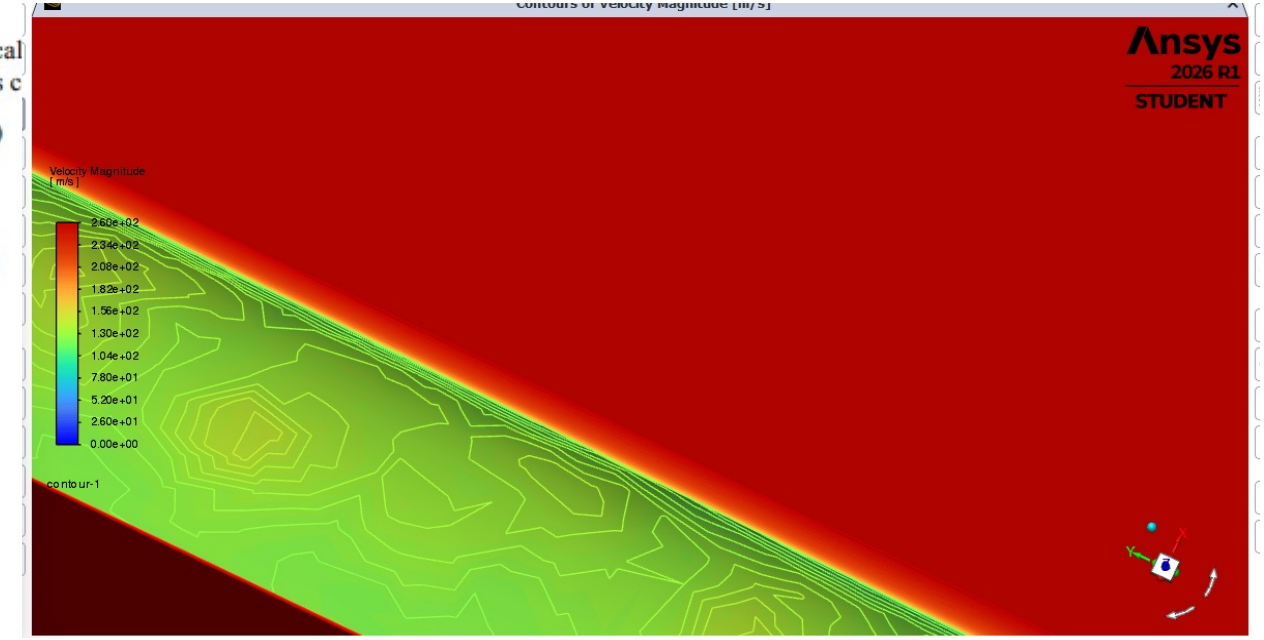
$$N_x = k_x \frac{\pi^2 D}{\ell^2}$$

Airbrakes Sizing Calculator

- Ran an ansys sim to account for boundary layer
- 1/7th power law for velocity profile in the boundary layer calculated
- Compared required drag using 10% of overall Cd to the boundary layer drag using the boundary layer height
- Rearranged drag equations to solve for the length, width was maintained

Max Actuation Value by Matlab Sizing Calculator: 5.56 mm





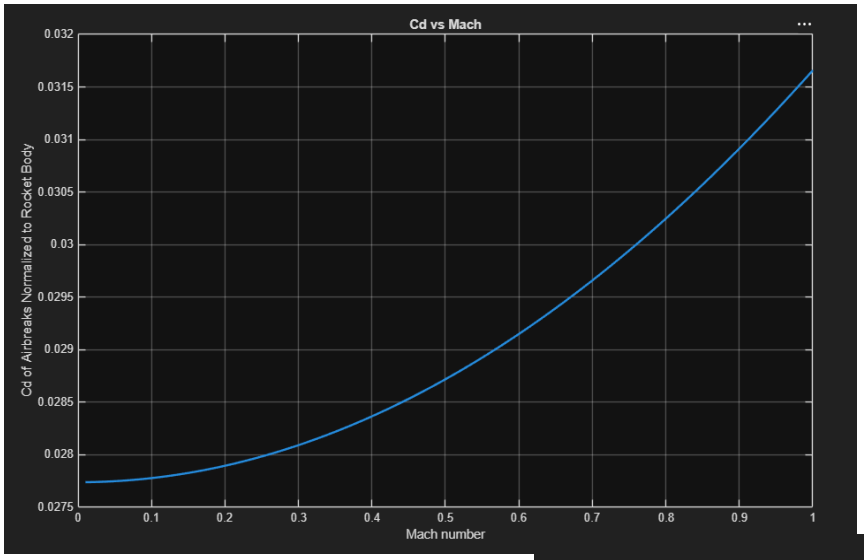
1/7th power law velocity profile for flow over a flat plate with no pressure gradient:

$$\frac{u}{U} = \begin{cases} \left(\frac{y}{\delta}\right)^{1/7} & 0 \le \frac{y}{\delta} < 1 \\ 1 & \frac{y}{\delta} \ge 1 \end{cases}$$

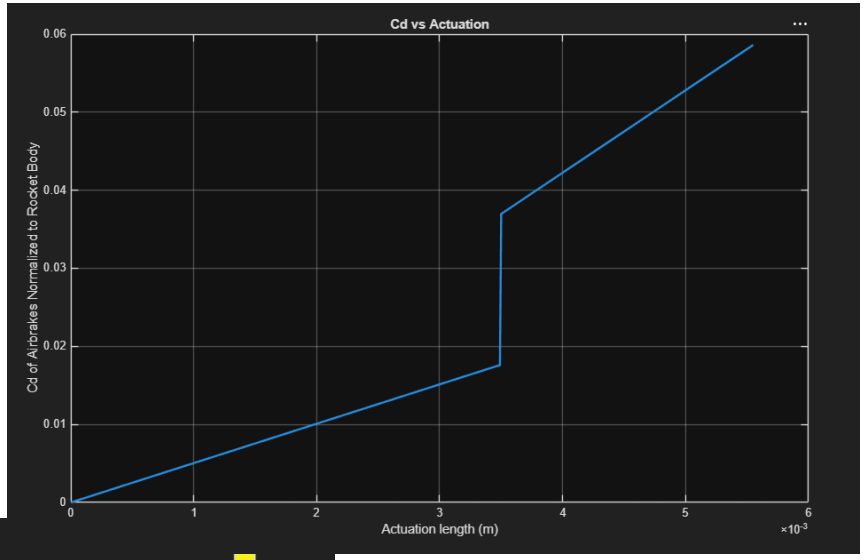
u = local flow velocity inside boundary layer
U = free stream velocity outside boundary layer
Y = normal distance out from rocket body
S = height of boundary layer

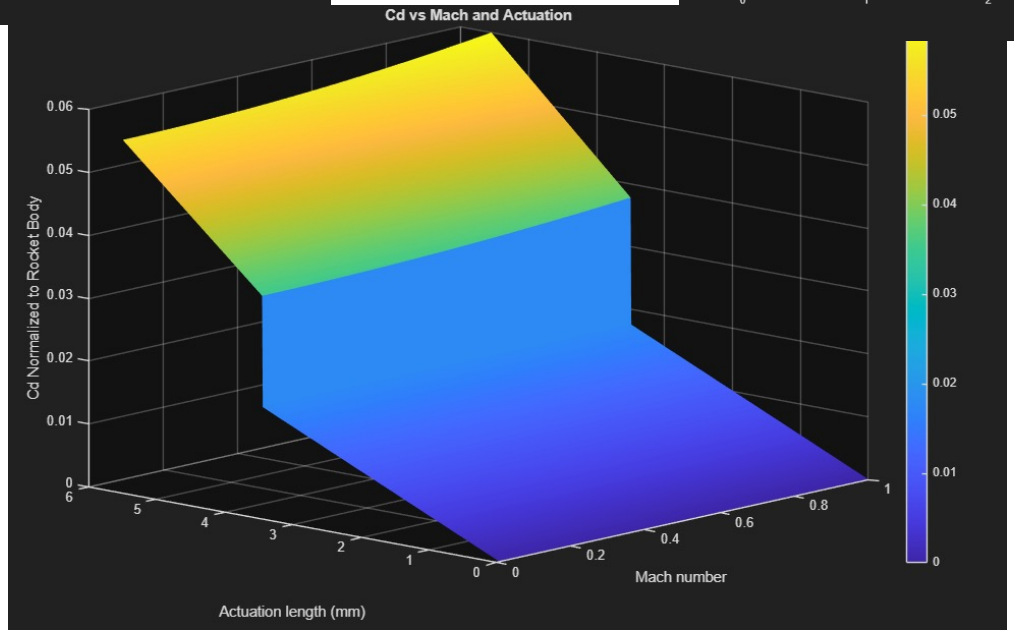
Airbrakes Sizing Calculator Graphs

- Used equations of Cd from OpenRocket Documentation (talk about that later) for constant actuation, Cd vs Mach graph shows a gentle increase as base drag is the only determining factor.
- Used the height of the boundary layer, and the average velocity in the boundary layer that was solved, to create an if statement where the Openrocket Documentation drag equations work with that boundary layer equation if the actuation is inside the boundary layer and combines the two if the airbrakes reach outside.



Can see the gentle increase with Base drag only.

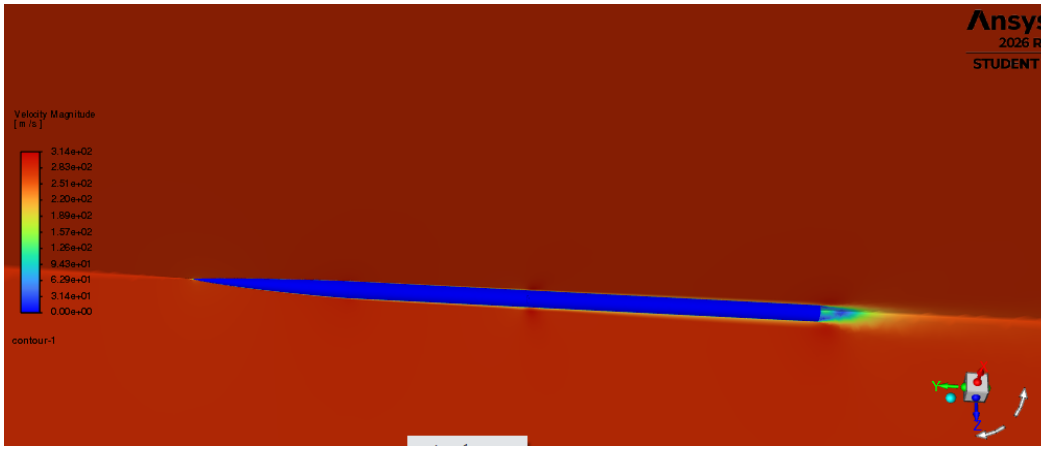




Can see the jump from when the airbrakes are in the boundary layer and outside of it.

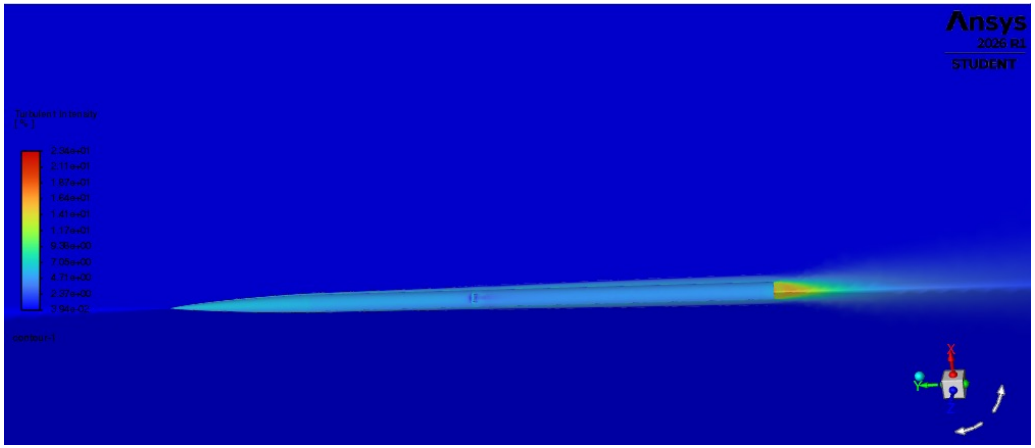
CFD Verifications

Max Mach, Max Actuation:
Cell Count: 116837
Cell Skewness Max: 0.49874826
Cell Orthogonality Min: 0.2009599
Airbrakes Cd Value: 0.10132213
Wall y+: 43.93292
Percent Error: 42.4% (estimate .0583316)



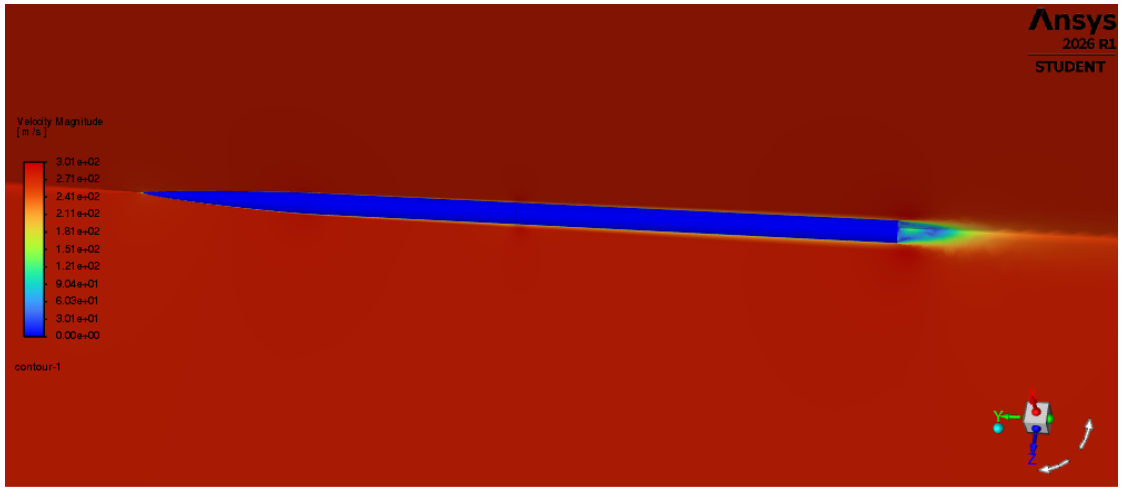
Similar results to Barrowmans Method, where higher mach induced higher percent error from estimated.

Max Actuation, Mach .3:
Cell Count: 114318
Cell Skewness Max: 0.49874826
Cell Orthogonality Min: 0.20051355
Airbrakes Cd Value: 0.082904796
Wall y+: 44.688452
Percent Error: 27% (estimate .055102)



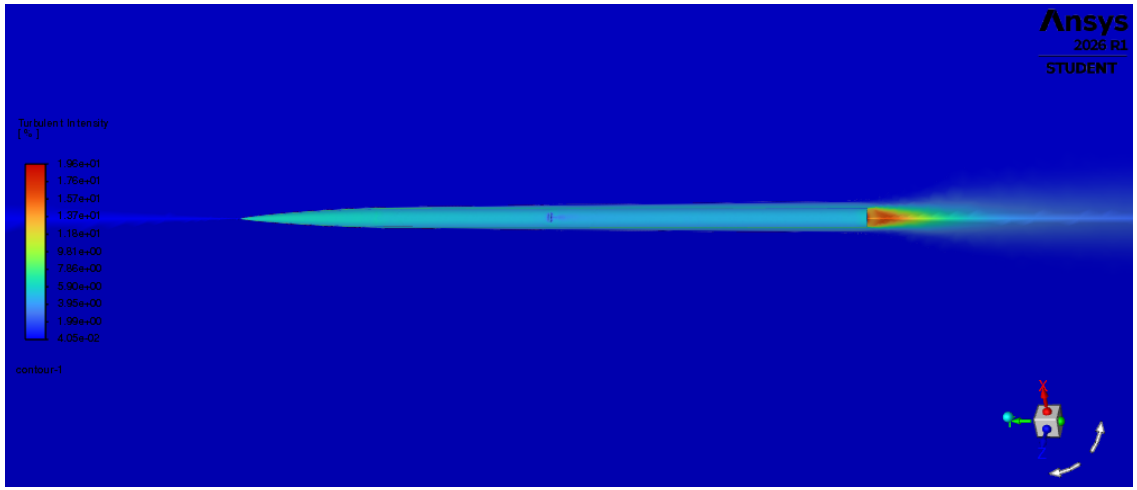
CFD Verifications

Airbrakes Half of Max Actuation at Max Mach (.816):
Cell Count: 115455
Cell Skewness Max: 0.53537905
Cell Orthogonality Min: 0.20064379
Airbrakes Cd Value: 0.039151422
Wall y+: 44.653041
Percent Error: 64% (Estimated .0141221)



Possible reason of extremely large error in both could be the Matlab calculations average velocity in the boundary layer, as half of max actuation sits inside the calculated boundary layer and innacurate drag equations used for boundary layers.

Airbrakes Half of Max Actuation at Mach .3:
Cell Count: 112878
Cell Skewness Max: 0.53537905
Cell Orthogonality Min: 0.20101081
Airbrakes Cd Value: 0.0322358
Wall y+: 47.554506
Percent Error: 57% (Estimated .0138195)



Full rocket Cd

- Used OpenRocket Documentation equations to find Pressure drag, Skin Friction drag, and Base Drag

- Note: Same equations used for Airbrake Sizing Calculator Pressure, Skin Friction, and Base drag.

Skin Friction Equations:

$$C_f = 0.032 \left(\frac{R_s}{L}\right)^{0.2}. \qquad (3.80)$$

Local Critical R:

$$R_{\text{crit}} = \frac{v_0\, x}{\nu},$$

$$C_f = \frac{1}{(1.50\ \ln R - 5.6)^2}. \qquad (3.78)$$

Altogether, the skin friction coefficient for turbulent flow is calculated by

$$C_f = \begin{cases} 1.48 \cdot 10^{-2}, & \text{if } R < 10^4 \\ \text{Eq. (3.78)}, & \text{if } 10^4 < R < R_{\text{crit}} \\ \text{Eq. (3.80)}, & \text{if } R > R_{\text{crit}} \end{cases}. \qquad (3.81)$$

A subsonic speeds the skin friction coefficient turbulent and roughness-limited boundary layers need to be corrected for compressibility with the factor

$$C_{f_c} = C_f\,(1 - 0.1\,M^2). \qquad (3.82)$$

In supersonic flow, the turbulent skin friction coefficient must be corrected with

$$C_{f_c} = \frac{C_f}{(1 + 0.15\,M^2)^{0.58}} \qquad (3.83)$$

$$(C_D)_{\text{friction}} = C_{f_c} \frac{\left(1 + \frac{1}{2f_B}\right) \cdot A_{\text{wet,body}} + \left(1 + \frac{2t}{\bar{c}}\right) \cdot A_{\text{wet,fins}}}{A_{\text{ref}}} \qquad (3.85)$$

This is why airbrakes have extremely low/negligible skin friction.

Base Drag Equations:

$$(C_{D\bullet})_{\text{base}} = \begin{cases} 0.12 + 0.13M^2, & \text{if } M < 1 \\ 0.25/M, & \text{if } M > 1 \end{cases} . \qquad (3.94)$$

Varies with the square of the Mach number in our rocket speed range and relates to the fin Pressure Drag.

Pressure Drag Equations:

Nose:

The nose pressure drag is approximately proportional to the square of the sine of the joint angle ϕ (shown in Figure 3.11) [25, p. 237]:

$$(C_{D\bullet,M=0})_p = 0.8 \cdot \sin^2 \phi. \tag{3.86}$$

This is why the von Karman nosecone is ideal; the joint angle is 0.

Fins:

$$(C_{D\bullet})_{LE\perp} = \begin{cases} (1-M^2)^{-0.417} - 1 & \text{for } M < 0.9 \\ 1 - 1.785(M - 0.9) & \text{for } 0.9 < M < 1 \\ 1.214 - \frac{0.502}{M^2} + \frac{0.1095}{M^4} & \text{for } M > 1 \end{cases}. \tag{3.89}$$

$$(C_{D\bullet})_{LE} = (C_{D\bullet})_{LE\perp} \cdot \cos^2 \Gamma_L \tag{3.91}$$

This is why the Clipped Delta is ideal; the sweep angle is higher than most fin designs.

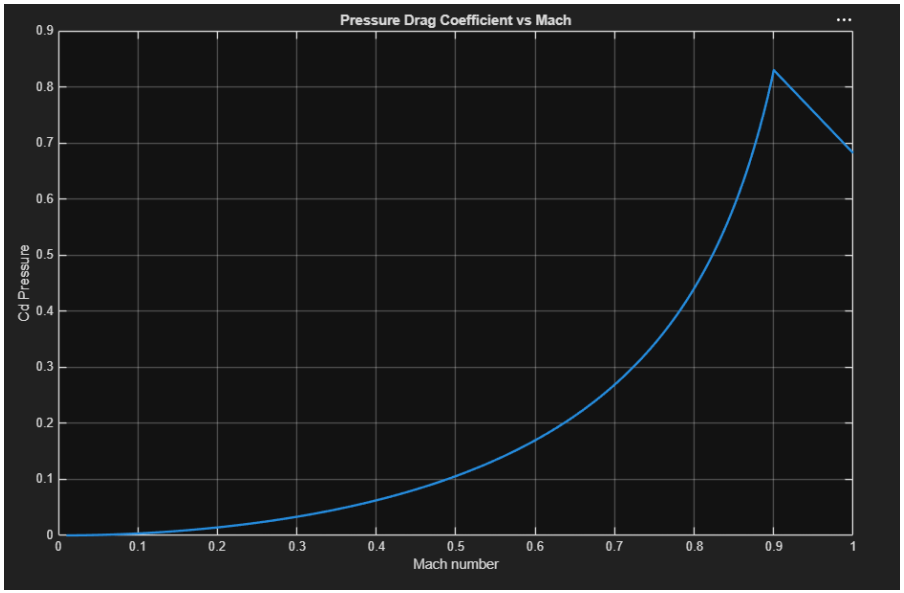
The fin base drag coefficient of a square profile fin is the same as the body base drag coefficient in equation 3.94:

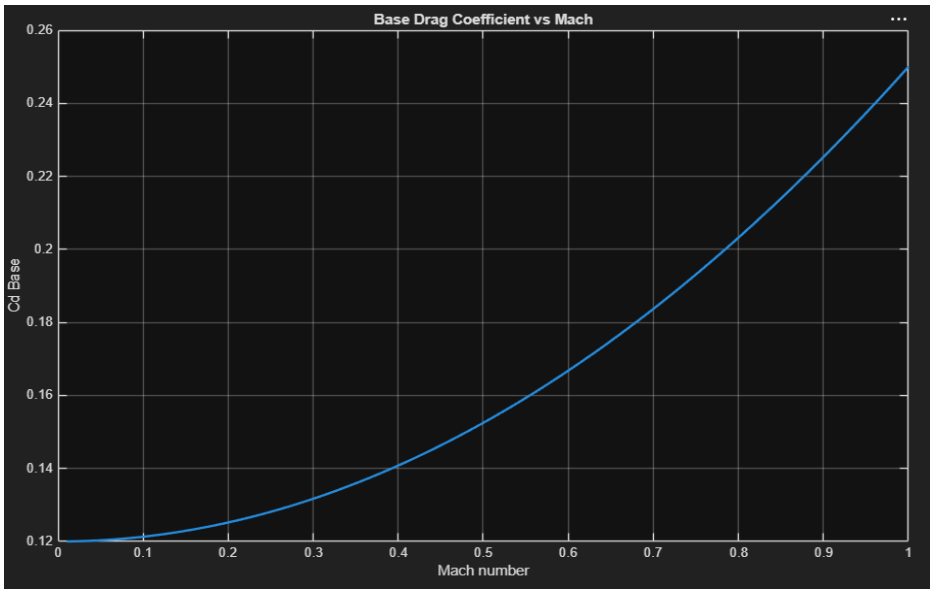
$$(C_{D\bullet})_{TE} = (C_{D\bullet})_{\text{base}} \tag{3.92}$$

The total fin pressure drag is the sum of the leading and trailing edge drags

$$(C_{D\bullet})_{\text{pressure}} = (C_{D\bullet})_{LE} + (C_{D\bullet})_{TE}. \tag{3.93}$$

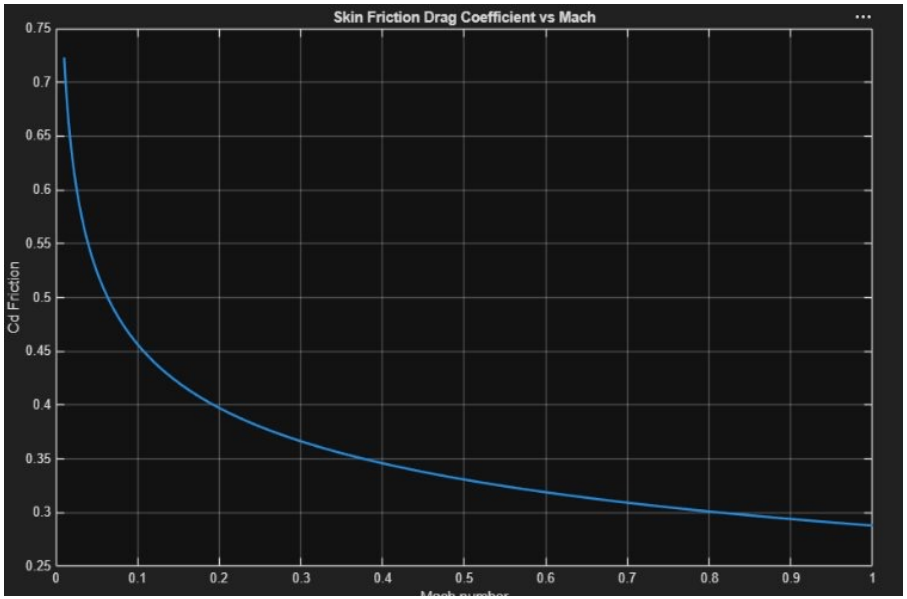
Full rocket Cd and CFD Verficiation



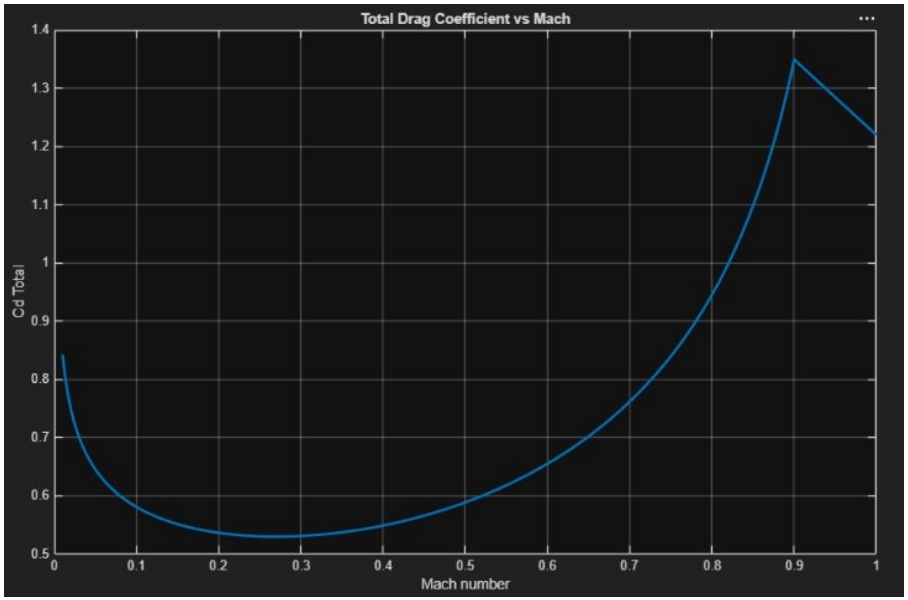


$$(C_{D\bullet})_{LE\perp} = \begin{cases} (1-M^2)^{-0.417} - 1 & \text{for } M < 0.9 \\ 1 - 1.785(M-0.9) & \text{for } 0.9 < M < 1 \\ 1.214 - \frac{0.502}{M^2} + \frac{0.1095}{M^4} & \text{for } M > 1 \end{cases} . \qquad (3.89)$$

$$(C_{D\bullet})_{\text{base}} = \begin{cases} 0.12 + 0.13M^2, & \text{if } M < 1 \\ 0.25/M, & \text{if } M > 1 \end{cases} .$$



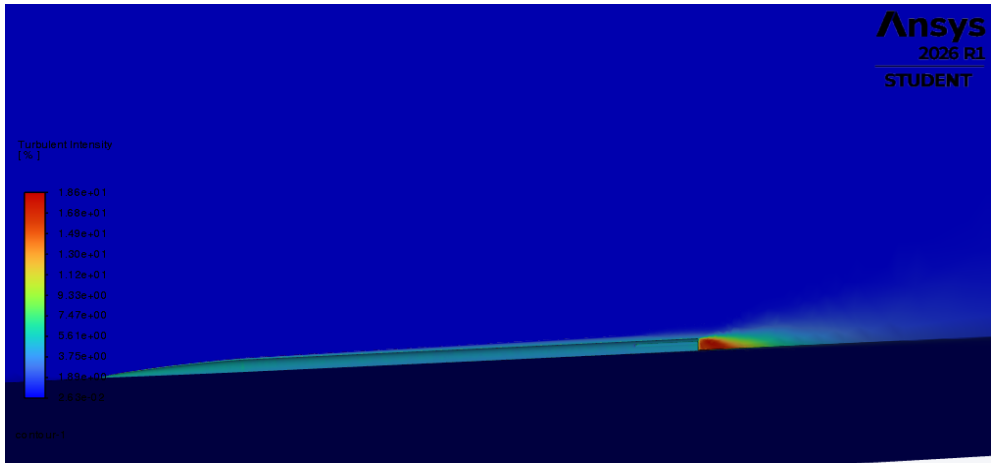
$$C_{f_c} = C_f\,(1 - 0.1\,M^2). \qquad (3.82)$$



Extreme static drag/skin friction decline 0-> .1
Base drag + Pressure Drag large increases .1 -> .9
New Equation for Leading Edge of Fins .9->1

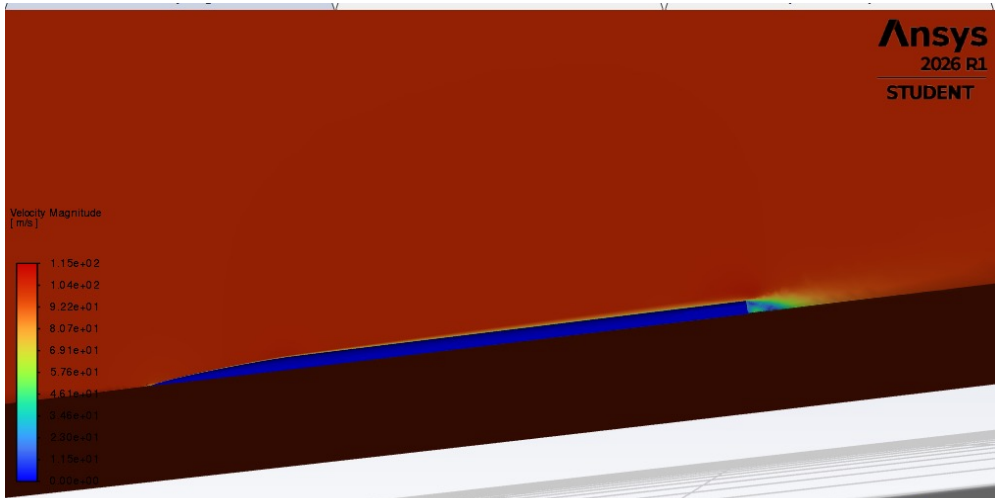
CFD Verifications

Full Body Drag Mach .3:
Cell Count: 193222
Cell Skewness Max: 0.66545541
Cell Orthagonality Min: 0.20006052
Cd = .5202;
Wall y+: 49.047418
Percent Error: 0% (Estimated .52)



Stairstepping did occur in two locations on the back face for both of these simulations, results were still taken as the back face has flow separation which means the boundary layer flow is separated and doesn't affect the drag calculations very much. This can be seen in the error, as the Cd graph was nearly exactly the result from .3 Mach but, similar to the other CFD, very different at the transonic range.

Full Body Drag Max Speed 280 (m/s):
Cell Count: 195378
Cell Skewness: 0.031316868
Cell Orthagonality: 0.88077855
Cd = .5239
Wall y+ : 48.697505
Percent Error: 81% (Estimated .95)



Compressibility Incorrect Calculations

- Barrowman's Method, Flutter Calculations, Wall Thickness Calculations, Airbrake Sizing Calculator, and Full Rocket Cd don’t accurately account for changes in either pressure, density, Cp changes, shocks or heat transfer.
- Even more accurate methods like the airbrake sizing calculators 1/7th power law isn't completely accurate as that requires a fixed boundary layer value, the boundary layer size doesn’t change with speed.

Compressibility Calculator Additions

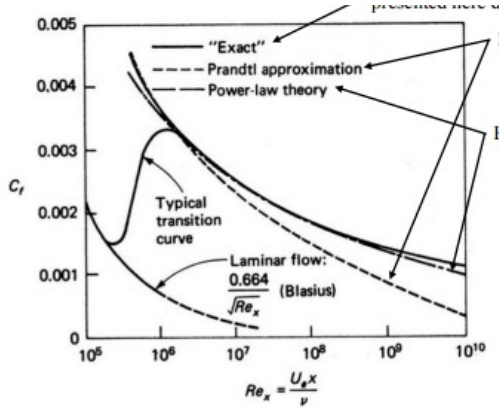
- Prandtl factor on fin lift: Compressibility increases the lift a fin makes. The fin CP also shifts rearward from the quarter chord assumptions we had before as Mach rises.
- Boundary-layer power law: Uses local Mach inside the boundary layer, since air near the wall is slower and hotter. Can change the 7 value to fit the compressibility equations from Ansys or some other CFD.

$$\beta = \sqrt{|M^2 - 1|} = \begin{cases} \sqrt{1 - M^2}, & \text{if } M < 1 \\ \sqrt{M^2 - 1}, & \text{if } M > 1 \end{cases} \quad (3.13)$$

$$P = \frac{1}{\beta} = \frac{1}{\sqrt{1 - M^2}} \quad (3.14)$$

1/7th power law velocity profile for flow over a flat plate with no pressure gradient:

$$\frac{u}{U} = \begin{cases} \left(\frac{y}{\delta}\right)^{1/7} & 0 \le \frac{y}{\delta} < 1 \\ 1 & \frac{y}{\delta} \ge 1 \end{cases}$$



Appendix

Vehicle Engineering

Bill of Materials – Airframe and Bulkheads

Item	Specification / BOM basis	Qty	Material	Source · Part #	Unit (g)	Qty × g	Line cost
Nose Cone	Von Kármán, 76.2 mm OD, 406 mm long	1	G12 Fiberglass	McMaster(sheet) · 8667K242	133	133	$44.56
Upper airframe	76.2 mm OD, 1 mm wall, 760 mm. Cut from 10 ft tube.	1	G12 Fiberglass	McMaster · 8535K74	332.2	332.2	$356.50
Bottom/Drogue bay airframe	76.2 mm OD, 1 mm wall, 690 mm. Cut from upper tube stock.	1	G12 Fiberglass	Same tube as upper · Same tube as upper	301.6	301.6	N/A
Airframe Coupler	Connects upper and lower airframe	1	G12 Fiberglass	McMaster · 8535K74	81.3	81.3	$55.00
Fins	Mass included in lower airframe per workbook	4	G10 Fiberglass	McMaster · 85345K112	Part of lower airframe	included	$11.26
Avionics Bay Bulkhead	Seals the ends of each side of the avionics bay, holds eye bolt	2	6061-T6 Al	Speedy Metals · N/A	127.13	254.26	$39.85
Drogue Bulkhead	Seals against ejection gas, holds eye bolt for shock cord	1	6061-T6 Al	Speedy Metals · N/A	127.13	127.13	N/A
Nose cone bulk-head	Caps the nose shoulder, holds eye bolt for shock cord	1	6061-T6 Al	Speedy Metals · N/A	127.13	127.13	N/A
Bulkhead Coupler	Bulkhead retaining sleeve	1	6061-T6 Al	Speedy Metals · N/A	36.4	36.4	N/A
Av bay	Holds controls	1	G12 Fiberglass	McMaster(sheet) · N/A	122.1	122.1	N/A
Airframe / bulkheads						**1,515.12**	**$507.17**

Bill of Materials – Thrust Structure and Propulsion

Item	Specification / BOM basis	Qty	Material	Source · Part #	Unit (g)	Qty × g	Line cost
Motor mount	K375NW description in BOM. Motor-tube mass corrected by team to 298.5 g.	1	G12 Fiberglass	Wildman Rocketry · G12-2.1	298.5	298.5	$52.27
Centering ring	Centers motor tube and distributes load	3	Birch	Makerstock custom cut · N/A	18.5	55.5	$69.75
Flame-Retardant Rod	One purchased stock rod yields 12 pins. Workbook lists 3 g stock mass.	1	Garolite	McMaster · 8669K623	3	3	$8.50
Retaining Ring+Cap	Keeps booster from falling out	1	Aluminum	Aeropack · 24067	39.7	39.7	$37.97
JB-weld	Two packages. Workbook lists 22.5 g per unit.	2	Epoxy	JB-Weld · 8265S	22.5	45	$15.98
Thrust structure						**441.7**	**$184.47**
Motor	CTI 2372K1440-17A	1	APCP	Wildman Rocketry · K1440	1893	1,893	$273.51
Propulsion						**1,893**	**$273.51**

Bill of Materials – Airbrakes

Item	Specification / BOM basis	Qty	Material	Source · Part #	Unit (g)	Qty × g	Line cost
Brake Panel	30 × 30 × 5 mm panel, as listed in BOM	4	AISI 1020 steel	McMaster · 89015K172	18.78	75.12	$11.32
Brake Bulkhead	Base plate with panel slits	2	6061-T6 Al	McMaster(sheet) · N/A	65.45	130.9	N/A
Brake Shaft	Two-arm horn, r = 16.72 mm	2	6061-T6 Al	McMaster(sheet) · N/A	2.91	5.82	N/A
Link	Horn-panel link	4	6061-T6 Al	McMaster(sheet) · N/A	0.43	1.72	N/A
Pin	Connects links	8	6061-T6 Al	McMaster(sheet) · N/A	0.04	0.32	N/A
Stepper Motor Mount	Retains stepper motor	1	6061-T6 Al	McMaster(sheet) · N/A	21.32	21.32	N/A
Airbrake threaded rod	Connect airbrake layers	1	AISI 1020 steel	Mcmaster · 93805A312	3.2	3.2	$9.51
Geared Stepper Motor	NEMA 17, 560 inozf, 5mm D-shaft	1	n/a	McMaster · 4801N12	1410.7	1,410.7	$352.36
PTFE Dry-Lubricant	Dry PTFE lubricant	1	PTFE	McMaster · 1130K15	~0	≈0	$14.70
Epoxy	Adhesive for bonded components	1	Epoxy	Aircraft Spruce · ES6209	0	0	$82.75
Airbrakes						**1,649.1**	**$470.64**

Bill of Materials – Recovery

Item	Specification / BOM basis	Qty	Material	Source · Part #	Unit (g)	Qty × g	Line cost
Main parachute	Annular D_0 1.65 m, 7 m/s, 1,075 N design load	1	Ripstop nylon, nylon thread	Rocketman · Rocketman Ultra-Light Annular, 5ft	176	176	$70.00
Drogue parachute	Cross D_0 0.43 m, 31.5 m/s. BOM opening load: 73 N (unresolved).	1	Ripstop nylon	N/A · N/A	20	20	$20.00
Main shock cord	Nose to upper airframe, 20 ft, 8,450 N	1	Tubular kevlar	LOC Precisicion · KV-25-30	40-50	40–50	$24.77
Drogue shock cord	Upper to lower airframe, 20 ft. BOM says 82,450 N (unconfirmed).	1	Tubular Kevlar	LOC Precisicion · KV-25-30	40-50	40–50	$24.77
Quick links	Six quick links, 4,003.4 N stated in BOM	6	Steel	McMaster · 8947T25	6	36	$6.66
Swivels	Six swivels. Material and mass as listed in workbook.	6	Plastic, aluminium	McMaster · 8928T41	77	462	$76.68
Eye bolts	Four bulkhead anchors	4	Zinc-plated steel	McMaster · 3032T16	110	440	$26.72

Bill of Materials – Recovery (cont.) and Fasteners

Item	Specification / BOM basis	Qty	Material	Source · Part #	Unit (g)	Qty × g	Line cost
Ejection charges	Two charges. BOM: 1.52 g main / 0.43 g drogue. Powder price ≈$30/lb.	2	Black powder	Launch club · FFFFg black powder	.94-1.6	1.88–3.2	TBD
E-matches	Two initiators, one per listed charge	2	N/A	MJG Tech · Fire Wire Initiator	3	6	≈$4.00
Shear pins	Four nylon shear pins for separation joints	4	Nylon	McMaster · 94270A120	0.11	0.44	$16.58
Shear pin hex nuts	Brass shear-pin nuts, bonded inside airframe	4	Brass	McMaster · 93187A100	0.03	0.12	$8.28
Mil Kapton Tape	Electrical insulation at charge wells	1	Silicone adhesive	Packaging Supplies · Y11675	0	0	$52.80
Permatex Silicone Gasket	Seals charge-system openings	1	Silicone	Amazon · 81409	0	0	$28.73
Recovery, known cost*						**1,222.44–1,243.76**	**$359.99**
Bulkhead bolt	Secures bulkhead to pressure and shock loads	4	18-8 Stainless Steel	McMaster · 97654A663	0.423	1.69	$8.92
Bulkheads bolt spacers	Secures Bolt	4	Aluminum	McMaster · 94669A079	0.5609	2.24	$7.48
Thumb screw	For av bay	1	18-8 Stainless Steel	McMaster · 92625A611	42.9	42.9	$13.60
Thumb screw spacer	Secures thumb screw	1	Aluminum	McMaster · 94669A359	0.6	0.6	$2.74
Fasteners						**47.44**	**$32.74**

Total Cost: ~$1828.52

Appendix – Vehicle Engineering Hand Calcs

#	Calculation	Slides
A1	Thrust structure load case and requirements	3, 4, 11
A2	Pin and motor tube wall checks	8, 11, 12
A3	Bond, ring, airframe and retainer checks	4, 9, 12
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#	Calculation	Slides
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A19	Airbrake pin, link and crank checks	45, 50, 53
A20	Airbrake drive shaft and coupling rod	44, 45, 53
A21	Inputs, allowables and sources	All

Convention: FOS = allowable ÷ applied. Required FOS ≥ 2.0 on structure (TS.1) and on parachute loads (FO.7.10).

Thrust Structure – Load Case and Requirements

CTI 2372K1440	Value
Peak thrust, T	1,828.9 N
Average thrust	1,437 N
Total impulse	2,372 N·s
Burn time	1.7 s
Loaded mass	1,893 g
Propellant	1,129 g
Spent motor	764 g
Size	54 × 572 mm

Item	Equation	Values	Result
FO.2.5 thrust-to-weight	$TWR = F_{avg} / (m \cdot g)$	1,437 / (6.98 × 9.81)	**21.0 ≥ 6**
Max liftoff mass, TWR 6	$m_{max} = F_{avg} / (6g)$	1,437 / (6 × 9.81)	**24.4 kg**
FO.8.11 total impulse	$I_t \leq 40{,}960$ N·s	2,372 N·s	**Met**
TS.2 motor fit, radial	$(D_i - D_{motor}) / 2$	(54.66 − 54) / 2	**0.33 mm**
Tube wall	$t = (D_o - D_i) / 2$	(57.81 − 54.66) / 2	**1.575 mm**
TS.3 unsupported span	Δx between rings	296.4 − 37.9; 555.0 − 296.4	**259 mm**
Worst-case pin load	$P = T / 4$	1,828.9 / 4	**457.2 N**
If all 12 pins share	$P = T / 12$	1,828.9 / 12	152.4 N

Worst case: hole clearance can leave one ring carrying all of T, so one ring's 4 pins each see T/4 = 457.2 N. Every check in A2–A3 uses this.

Thrust Structure – Pin and Tube Wall Checks

Check	Equation	Substituted values	Applied	Allowable	FOS
Pin shear, G10 (1 plane)	$\tau = P / (\pi d^2/4)$	457.2 / (π·10²/4) = 457.2 / 78.54	5.82 MPa	148 MPa	**25.4**
Pin bending, no ring support	$\sigma = 32PL / (\pi d^3)$	32 × 457.2 × 9.5 / (π × 10³)	44.2 MPa	345 MPa	**7.8**
Bearing, 1 mm airframe	$\sigma_{br} = P / (d \cdot t)$	457.2 / (10 × 1.0)	45.7 MPa	241 MPa	**5.3**
Bearing, 1.575 mm motor tube	$\sigma_{br} = P / (d \cdot t)$	457.2 / (10 × 1.575)	29.0 MPa	241 MPa	**8.3**
Tube shear-out, aft holes	$\tau = P / (2 \cdot e \cdot t)$	e = 37.9 − 5 = 32.9 mm; 457.2 / (2 × 32.9 × 1.575)	4.41 MPa	148 MPa	**33.5**
Old: birch plywood pins	$\tau = P / (\pi d^2/4)$	5.82 MPa vs 2.22 MPa planar shear	5.82 MPa	2.22 MPa	0.38 ✗
Old: aft pins at 25 mm	$\tau = P / (2 \cdot e \cdot t)$	e = 20 mm; 457.2 / (2 × 20 × 1.575)	7.26 MPa	148 MPa	20.4

Lowest pin-path check: bearing in the 1 mm airframe wall, FOS 5.3. The pins alone carry full thrust (TS.4).

Thrust Structure – Bond, Airframe and Retainer

Check	Equation	Substituted values	Applied	Allowable	FOS
Bond, ring inner face	$\tau = T / A_{in}$	1,828.9 / 3,103 mm^2 (CAD, skeletal ring)	0.589 MPa	4.83 MPa	**8.2**
Bond, ring outer face	$\tau = T / A_{out}$	1,828.9 / 4,132 mm^2	0.443 MPa	4.83 MPa	**10.9**
Ring shear, plywood	$\tau = T / A_{in}$	1,828.9 / 3,103 mm^2	0.589 MPa	9.5 MPa	**16.1**
Airframe compression	$\sigma = T / [\pi(D_o^2 - D_i^2)/4]$	1,828.9 / [π(76.2² − 74.2²)/4] = 1,828.9 / 236.2	7.74 MPa	241 MPa	**31.1**
Euler buckling, whole vehicle, pinned	$P_{cr} = \pi^2 EI / L^2$	I = π(76.2⁴ − 74.2⁴)/64 = 1.670×10⁵ mm^4; π² × 10,300 × 1.670×10⁵ / 1,907²	1,829 N	4,669 N	**2.55**
Retainer bond, hot, at deployment	$\tau = m \cdot n \cdot g / (\pi \cdot D \cdot L)$	0.764 × 50 × 9.81 / (π × 57.81 × 12.7) = 374.7 / 2,307	0.162 MPa	9.43 MPa	**58**

Lowest overall: Euler buckling, FOS 2.55. The bond alone carries full thrust at FOS 8.2 (TS.4).

Thrust Structure – Bond-Line Thermal Model

Input	Value	Basis
Case temperature, T_c	200 °C	Slide 9 boundary assumption
Initial temperature, T_0	40 °C	Hot launch day
Wall thickness, L	1.575 mm	G12 motor tube
Conductivity, k	0.61 W/m·K	NIST G-10
Specific heat, c_p	999 J/kg·K	NIST G-10
Density, ρ	1,850 kg/m^3	G12
Diffusivity, $\alpha = k/(\rho c_p)$	3.30×10^{-7} m^2/s	Calculated

With the assumed 200 °C case boundary, the model approaches 200 °C. Bond strength at temperature still requires verification.

$$T_{bond}(t) = T_c - (T_c - T_0)\,\Sigma\, c_n\, e^{-\lambda_n^2\, Fo}$$
$$c_n = 4(-1)^n / [(2n+1)\pi],\ \lambda_n = (2n+1)\pi/2$$
$$Fo = \alpha t / L^2$$

1-D conduction through the tube wall: case side held at T_c from ignition, bond side insulated (no heat escapes). Both choices push the bond line hotter, so this is an upper bound.

Time	Fo	Bond line	Meaning
1.1 s	0.144	60 °C	Std. epoxy weakens
1.7 s	0.226	84 °C	Motor burnout
5 s	0.665	161 °C	
10 s	1.33	192 °C	
60 s	7.98	200 °C	Soaked; < 260 °C

Recovery – Descent Rates and Deployment Timing

$$v = \sqrt{(2mg / (\rho \cdot C_D \cdot S_0))}, \quad S_0 = \pi D_0^2/4, \quad m = 5.85 \text{ kg (burnout)}$$

Stage	$C_D \cdot S_0$	Values	v	Requirement
Drogue, cross, sea level	$0.725 \times \pi(0.43)^2/4 = 0.105\ m^2$	$\sqrt{(2 \times 5.85 \times 9.81 / (1.225 \times 0.105))}$	29.8 m/s	15.2–45.7 m/s
Drogue, ρ = 1.10 kg/m³	$0.105\ m^2$	$\sqrt{(2 \times 5.85 \times 9.81 / (1.10 \times 0.105))}$	**31.5 m/s**	15.2–45.7 m/s ✓
Main, annular, sea level	$0.90 \times \pi(1.65)^2/4 = 1.924\ m^2$	$\sqrt{(2 \times 5.85 \times 9.81 / (1.225 \times 1.924))}$	**7.0 m/s**	≤ 9.1 m/s ✓
Main, ρ = 1.10 kg/m³	$1.924\ m^2$	$\sqrt{(2 \times 5.85 \times 9.81 / (1.10 \times 1.924))}$	7.4 m/s	≤ 9.1 m/s ✓
Earlier mass case, 4.9 kg	same	drogue / main at sea level	27.3 / 6.39	–

Timing check	Values	Result
Time from main deploy to 305 m	(450 − 305) m / 31.5 m/s	4.6 s to open and slow
FO.7.2.3, ⅓ of apogee	3158 m / 3 = 1,052.6 m; drogue only above, 31.5 > 15.2 m/s	Met

Recovery – Parachute Trade Study Scoring

Criterion (BDM)	Wins / 6	Weight
Opening force	3	50.0%
Descent velocity	2	33.3%
Packed size, D_0	1	16.7%
Mass	0	0%

$$S_{ij} = w_j \cdot (1/x_{ij}) \,/\, \Sigma_k\,(1/x_{kj})$$
$$\text{Total}_i = \Sigma_j\, S_{ij} \qquad \text{(lower x scores higher)}$$

Worked: annular opening force = 0.5 × (1/507) / (1/507 + 1/532 + 1/585) = 0.177 (rounded).

Main	Annular	Hemi	Flat
Opening force	0.177	0.169	0.154
Descent velocity	0.112	0.110	0.112
Packed size	0.059	0.053	0.055
Total	**0.348**	0.332	0.320

Drogue	Cross	Hemi	Flat
Opening force	0.206	0.152	0.142
Descent velocity	0.112	0.110	0.111
Packed size	0.055	0.055	0.057
Total	**0.373**	0.317	0.310

Selected: annular main (0.348) and cross drogue (0.373). Main opening forces on slide 25 are 507 / 532 / 585 N at 5.85 kg.

Recovery – Drogue and Main Opening Loads

$F = C_D S \cdot q \cdot C_x \cdot X_1$, $q = \frac{1}{2}\rho v^2$, $A = 2m / (C_D S \cdot \rho \cdot n \cdot D_0)$ (Pflanz method, Knacke)

Drogue case	v = g·t	q	F = 0.105 · q · 1.2 · 1.0	F
Nominal, apogee + 1 s	9.8 m/s	58.9 Pa	0.105 × 58.9 × 1.2	7.4 N
Backup, + 3 s	29.4 m/s	531 Pa	0.105 × 531 × 1.2	67.0 N
Late worst case, + 5 s	49.1 m/s	1,474 Pa	0.105 × 1,474 × 1.2	**186 N**
Drogue design load			Slide 27 reported load*	**394 N***

Main (annular)	Values	Result
$C_D S$ (load case, C_D 0.95)	0.95 × 2.138 m^2	2.031 m^2
Ballistic parameter A	2 × 5.85 / (2.031 × 1.225 × 8 × 1.65)	0.356 → $X_1 \approx 0.293$
q at drogue descent rate	½ × 1.225 × 31.5^2	608 Pa
Opening load	2.031 × 608 × 1.4 × 0.293	**507 N**
Main design load	507 × 2.12	**1,075 N**

Recovery – Snatch Loads and Pop-Test Limits

$$k = F_b / (\varepsilon \cdot L), \quad F_s = \Delta v \cdot \sqrt{(k \cdot m_{eff})}, \quad m_{eff} = m_1 m_2 / (m_1 + m_2), \quad \Delta v_{max} = 1{,}075\ \text{N} / \sqrt{(k \cdot m_{eff})}$$

Shock cord	Values	Result
Break strength, F_b	LOC KV-25-30, ¼ in tubular Kevlar	8,450 N
Elongation at break, ε	Kevlar braid 3–6%; 4% assumed	4%
Length, L	20 ft each	6.1 m
Stiffness, k	8,450 / (0.04 × 6.1)	**34.6 kN/m**

Joint	m_1 / m_2	m_{eff}	$\sqrt{(k \cdot m_{eff})}$	Δv for 1,075 N
Drogue	2.21 / 3.64 kg	2.21 × 3.64 / 5.85 = 1.375 kg	218.2 N·s/m	**1,075 / 218.2 = 4.9 m/s**
Nose (main)	0.50 / 5.35 kg	0.50 × 5.35 / 5.85 = 0.457 kg	125.8 N·s/m	**1,075 / 125.8 = 8.5 m/s**

Pop-test pass limits: separation ≤ 4.9 m/s at the drogue joint and ≤ 8.5 m/s at the nose keeps snatch at or below the 1,075 N design load.

Recovery – Link Load Check

FOS = rated load / design load: main chain 1,075 N, drogue chain 394 N

Part	Qty	Assumed rating	Main FOS (1,075 N)	Drogue FOS (394 N)
Shock cord, ¼ in Kevlar	2	8,450 N	7.9	21.4
Eye bolt, M10 316 SS	4	5,338 N	5.0	13.5
Quick link, assumed 800 lb	6	3,560 N	**3.3**	**9.0**
Swivel, new (1,100 lb)	6	4,893 N	4.6	12.4
Swivel, previous (300 lb)	–	1,334 N	1.24 ✗	3.4

FO.7.10: FOS ≥ 2 on drogue drag	Values	Result
Quick link ÷ late opening load	3,560 / 186	**19.1**
Same, on factored load	3,560 / 394	**9.0**

Recovery – Shear Pins and Ejection Charge Sizing

Step	Equation	Values	Drogue / Main
Pin strength, M3 nylon 6/6	$F_{pin} = \tau \cdot \pi d_3^2/4$	66 MPa × $\pi(2.387)^2/4$ = 66 × 4.48 mm^2; 2 pins per joint	295 N; 590 N/joint
FO.1.2.2 pressure held	$P = F_{joint} / A_{bore}$	590 N = 132.8 lbf; ÷ 6.70 in^2 (⌀74.2)	19.8 psi ≥ 15, MS 0.32
FO.1.2.3 target pressure	$P \geq 3F_{joint} / A$	6.35 in^2 (⌀72.2) = 28.23 N/psi; 3 × 590 / 28.23 = 62.7	**65 psi used**
Bay volume	$V_B = \pi D^2 L / 4$	D = 74.2 mm; L = 157 / 312 mm	680 / 1,351 cm^3
Free volume (CAD)	$V_F = V_B - V_{chute} - V_{hw}$	458 / 626 cm^3 = 27.95 / 38.20 in^3	458 / 626 cm^3
Primary charge	$m_p = PV / (RT)$	65 × 27.95 / (266 × 3,307) = 2.07×10^{-3} lbm	**0.94 / 1.28 g**
Backup charge	$m_b = 1.25\ m_p$	1.25 × 0.94; 1.25 × 1.28	**1.17 / 1.60 g**
Ejection force	$F_E = P \cdot A_{bore}$	65 psi × 6.70 in^2 = 435.6 lbf	**1,938 N**
FO.1.2.3 force ratio	$F_E / F_{joint} \geq 3$	1,938 / 590	**3.3**

Recovery – Pin Holes, Fasteners and Coupler

Check	Equation	Substituted values	Applied	Allowable	FOS
Wall bearing as pin breaks	$\sigma = F_{pin} / (d \cdot t)$	295 / (3 × 1.0)	98.5 MPa	241 MPa	**2.4**
Wall shear-out, 35 mm edge	$\tau = F_{pin} / (2 \cdot e \cdot t)$	295 / (2 × 35 × 1.0)	4.22 MPa	148 MPa	**35**
Nut fillet vs install torque	$T = \tau \cdot A \cdot r$	5.0 MPa × 28 mm^2 × 2.75 mm	0.10 N·m	0.39 N·m	**3.9**
M5 screw shear (fwd bulkhead)	$\tau = (F_E/4) / A_{minor}$	484 / (π × 4.019^2/4) = 484 / 12.69	38.2 MPa	309 MPa	**8.1**
⌀10 spacer bearing, 1 mm G12	$\sigma = (F_E/4) / (d \cdot t)$	484 / (10 × 1.0)	48.4 MPa	241 MPa	**5.0**
M5 thread strip, 6 mm engaged	$\tau = F / (½\pi D L_e)$	484 / (0.5π × 5 × 6)	10.3 MPa	139 MPa	**13.5**
Av bay thumb screw, ⌀10	$\sigma = m \cdot n \cdot g / A_s$	0.4 × 86 × 9.81 = 337 N; / 58.0 mm^2	5.8 MPa	205 MPa	**35**
FO.1.2.1 coupler engagement	$L / D_{body} \geq 1$	81 / 76.2 sep.; 110 / 76.2 bonded	1.06; 1.44	1.0 cal	**Met**

Recovery – Bulkhead and Bond Checks

Check	Equation	Substituted values	Applied	Allowable	FOS
Web bending at 65 psi (simply supported)	$\sigma = 3(3+\nu)pa^2 / (8t^2)$	$3 \times 3.33 \times 0.448 \times 32.1^2 / (8 \times 5.8^2)$	17.1 MPa	241 MPa	**14.1**
Hub under eye bolt, 1,075 N	$\sigma = 3W/(2\pi t^2)\cdot[(1+\nu)\ln(a/r_0)+1]$	$3 \times 1{,}075 / (2\pi \times 5.8^2) \times [1.33 \ln(32.1/10) + 1]$	38.9 MPa	241 MPa	**6.2**
M10 thread strip, 1,075 N	$\tau = W / (\tfrac{1}{2}\pi D L_e)$	1,075 / (0.5π × 10 × 15)	4.56 MPa	139 MPa	**30.5**
Rim bond, aft ⌀74.2, 130°F	$\tau = F_E / (\pi D h)$	1,938 / (π × 74.2 × 20)	**0.416 MPa**	1.2 MPa	**2.9**
Rim bond, ⌀72.2 drogue / nose, 130°F	$\tau = F_E / (\pi D h)$	1,835 / (π × 72.2 × 20)	0.405 MPa	1.2 MPa	**3.0**
Coupler bond, 110 mm, 130°F	$\tau = F_E / (\pi D L)$	1,938 / (π × 74.2 × 110)	0.076 MPa	1.2 MPa	**15.9**

Slide 33 baseline: aft rim-bond FOS 2.9. The calculation assumes a 5.8 mm web and the stated adhesive allowable.

Airbrakes – Linkage Kinematics

$$x(\theta) = r\cos\theta + \sqrt{(l^2 - r^2\sin^2\theta)}, \quad e = x(\theta) - x(55°), \quad r = 16.7\ \text{mm}, \ l = 13.7\ \text{mm}$$
$$dx/d\theta = -r\sin\theta - r^2\sin\theta\cos\theta \,/\, \sqrt{(l^2 - r^2\sin^2\theta)}$$

Check	Values	Result
Full out, θ = 0°	16.7 + 13.7	30.4 mm
Stowed, θ = 55°	$16.7\cos 55° + \sqrt{(13.7^2 - (16.7\sin 55°)^2)} = 9.58 + 0.74$	10.32 mm
Stroke, e_{max}	30.40 − 10.32	**20.1 mm**
Crank travel limit	$r\sin\theta \le l \rightarrow \text{asin}(13.7 / 16.7)$	55.1°

θ	e (mm)	|dx/dθ| (mm/rad)
0°	20.1	0
10°	19.5	6.5
20°	17.8	12.9
30°	15.0	19.5
40°	11.0	26.9
50°	5.3	40.8

Slide 45 targets 20.4 mm travel. The stated geometry gives 20.1 mm at 55°. Resolve this difference before freezing the mechanism.

Airbrakes – Panel Drag and Bending

Step	Equation	Values	Result
Dynamic pressure, M 0.995	$q = \frac{1}{2}\rho v^2$	63,070 Pa (slide 46 input)	63,070 Pa
Compressibility	$K_c = 1 + M^2/4 + M^4/40$	$1 + 0.995^2/4 + 0.995^4/40$	1.272
Corrected pressure	$q_c = K_c \cdot q$	1.272 × 63,070	80,230 Pa
Exposed area	$A = e \cdot w$	20.4 × 25 mm	5.1 cm^2
Drag per panel	$F = q_c \cdot C_d \cdot A$	$80{,}230 \times 1.28 \times 5.10\times10^{-4}$	**52.4 N**
All four panels	$F_{total} = 4F$	4 × 52.374	209.5 N
Root moment	$M = F \cdot e/2$	52.374 × 0.0102 m	0.534 N·m
Section inertia	$I = bh^3/12$	$25 \times 5^3 / 12 = 260.4\ mm^4$	$2.60\times10^{-10}\ m^4$
Bending stress	$\sigma = Mc/I$	$0.534 \times 0.0025 / 2.604\times10^{-10}$	5.13 MPa
Panel FOS, AISI 1020	σ_y / σ	210 / 5.13	**41**

Airbrakes – Track Friction and Motor Torque

Step	Equation	Values	Result
Inner reaction	$R_{in} = F \cdot a / L$	52.4 × 10.2 / 6.5	82.2 N
Outer reaction	$R_{out} = F + R_{in}$	52.4 + 82.2	134.6 N
Friction per panel	$F_f = \mu(R_{in} + R_{out})$	0.10 × 216.8	**21.7 N**
Track bearing	$\sigma = R_{out} / A_c$	134.6 / 15 mm^2	8.97 MPa
Track FOS, 6061-T6	σ_y / σ	241 / 8.97	**27**
Retract torque	$T(\theta) = 4 \cdot F_f(\theta) \cdot \lvert dx/d\theta \rvert$	F ∝ e, so F_f falls as panels retract	at each θ
Peak, θ ≈ 29°	$e = 15.4\ mm$	4 × 13.6 N × 0.0186 m	**1.01 N·m**
Holding-torque ratio	T_{motor} / T_{peak}	3.95 / 1.01	**3.9**

θ	T (N·m)
0°	0
10°	0.53
20°	0.90
29°	**1.01**
40°	0.83
50°	0.41
55°	0

Airbrakes – Track Plate Bond

ES6209 condition	Lap shear	Knockdown	Allowable
77°F	20.0 MPa	25%	5.0 MPa
130°F	4.8 MPa	25%	1.2 MPa

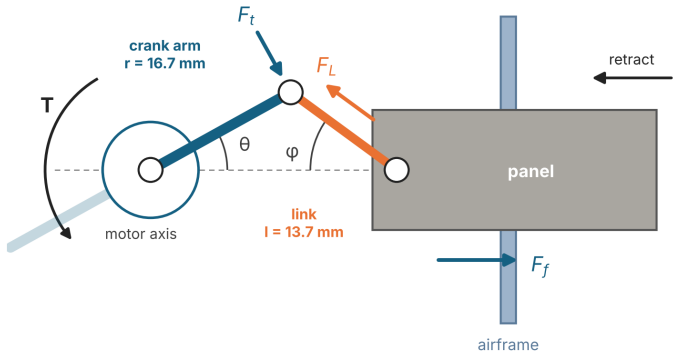
$$\tau = F / A_{bond}$$
$$A_{bond} = 41.5\ cm^2$$

Load case	F	Values	τ	FOS 77°F	FOS 130°F
Ejection, 65 psi	1,938 N	1,938 / 4,150 mm^2	0.467 MPa	10.7	**2.6**
Main opening design load	1,075 N	1,075 / 4,150 mm^2	0.259 MPa	19.3	4.6

Lowest airbrake FOS: 2.6 at 130°F under the 65 psi ejection charge, above the 2.0 requirement.

Airbrakes – Linkage Load Cases

Part (CAD)	Geometry	Material
Crank arm ×4	r = 16.72 mm; 5 × 2 mm arm	6061-T6
Link ×4	13.6 mm pin to pin; 5 × 2 mm	6061-T6
Pin ×8	Ø2.5 × 3 mm; 2 mm + 1 mm engaged	6061-T6
Panel pin hole	Ø2.5 × 1 mm deep; 2.5 mm to inner edge	AISI 1020
Coupling rod	6 mm threaded rod; 5 mm into each crank	18-8 SS
Motor shaft	Ø5 mm D-shaft; 5 mm into the lower crank	Steel



Case	Equation	Values	Result
Flight: friction, full out	$F_L = F_f / \cos\varphi$	21.7 / cos 0°	**21.7 N**
Flight: torque peak, θ ≈ 29°	$F_t = T_{peak} / (4r)$	1.01 / (4 × 0.0167)	**15.1 N**
Jam: stall into one arm	$F_t = T_{motor} / r$	3.95 / 0.0167	**237 N**
Jam at θ = 29°, ϕ = 36.2°	$F_L = T_{motor} / (r \cdot \sin(\theta + \varphi))$	3.95 / (0.0167 × 0.908)	**261 N**

θ	5°	10°	20°	29°	40°	50°
Jam F_L (N)	1,229	625	337	261	237	271

Flight loads are small: the link carries at most 21.7 N. A panel jammed at motor stall is the sizing case, with 12× the link force and 16× the arm force.

Airbrakes – Pin, Link and Crank Checks

Check	Equation	Values	Capacity	FOS flight	FOS jam
Pin shear, one plane	$\tau_y \cdot \pi d^2/4$	139 × 4.91 mm^2	683 N	31.5	2.62
Pin bending, 1 mm side	$\sigma_y \cdot (\pi d^3/32) / a$	241 × 1.534 / 0.75	493 N	22.7	**1.89**
Panel hole bearing, 1 mm	$\sigma_y \cdot d \cdot t$	210 × 2.5 × 1	525 N	24.2	2.02
Link blind-hole bearing	$\sigma_y \cdot d \cdot t$	241 × 2.5 × 1	603 N	27.8	2.31
Link net section	$\sigma_y \cdot (w - d) \cdot t$	241 × 2.5 × 2	1,205 N	55.5	4.63
Link tear-out, blind end	$\tau_y \cdot 2t(e - d/2)$	139 × 2 × 1 × 1.25	348 N	16.0	**1.33**
Panel tear-out, inner edge	$\tau_y \cdot 2t(e - d/2)$	121 × 2 × 1 × 1.25	303 N	14.0	**1.16**
Crank arm tear-out	$\tau_y \cdot 2t(e - d/2)$	139 × 2 × 2 × 1.25	695 N	32.0	2.67
Link buckling, deploy	$\pi^2 EI / l^2$	π^2 × 68,900 × 3.33 / 13.6^2	12,255 N	564.8	47.0
Crank arm bending at root	$\sigma_y \cdot (tw^2/6) / (r - 5.28)$	241 × 8.33 / 11.44	176 N (F_t)	**11.6**	**0.74**

Flight passes, a jam does not: lowest flight FOS is 11.6 (crank arm). At full stall the arm yields (0.74). Limiting motor output to 1.47 N·m keeps every linkage part at FOS ≥ 2 in a jam.

Airbrakes – Drive Shaft and Coupling Rod

Check	Equation	Values	Capacity	Flight T	FOS flight	FOS jam
Rod torsion, M6 (d_3 4.77)	$T = \tau_y \cdot \pi d_3^3/16$	118 × 21.4 mm^3	2.53 N·m	0.505 N·m	5.00	**0.64**
Rod-to-crank bond, 77°F	$T = \tau \cdot \pi dL \cdot d/2$	5.0 × π·6·5 × 3	1.41 N·m	0.505 N·m	2.80	**0.36**
Rod-to-crank bond, 130°F	$T = \tau \cdot \pi dL \cdot d/2$	1.2 × 94.2 × 3	0.34 N·m	0.505 N·m	**0.67**	**0.09**
Shaft-to-crank bond, 77°F	$T = \tau \cdot \pi dL \cdot d/2$	5.0 × π·5·5 × 2.5	0.98 N·m	1.01 N·m	**0.97**	**0.25**
Shaft-to-crank bond, 130°F	$T = \tau \cdot \pi dL \cdot d/2$	1.2 × 78.5 × 2.5	0.24 N·m	1.01 N·m	**0.23**	**0.06**
D-flat bearing on the hub	$T = \sigma_y \cdot d \cdot h \cdot L/2$	241 × 5 × 0.5 × 5 / 2	1.51 N·m	1.01 N·m	**1.49**	**0.38**

The torque path is the weak point: the CAD shows plain round bores at the shaft and the rod. A bond alone fails, and 5 mm of D-flat engagement gives FOS 1.49 in flight.

Fix: D-bore with a set screw and ≥ 6.7 mm of shaft engagement (FOS 2 on the flat), plus a cross pin or set screw through each crank hub into the rod. Re-run with the final hub detail.

Appendix – Inputs, Allowables and Sources

Input	Value used	Source
Motor, CTI 2372K1440	1,828.9 N peak, 1,437 N avg, 1,893 / 1,129 g	ThrustCurve.org
Fiberglass (G12 tube, G10 pins)	241 bearing/comp., 148 shear, 345 flex. MPa; E 10.3 GPa	Acculam, Gund, Trident G10/FR4 data (low end)
Birch plywood	9.5 MPa panel shear; 2.22 MPa planar shear	Metsä Wood birch plywood DoP
J-B Weld Original	7.17 MPa lap shear; 9.43 MPa after 3 h at 204 °C; 260 °C	J-B Weld TDS JBWTT0001
Aeropoxy ES6209	20.0 MPa at 77°F, 4.8 MPa at 130°F; 25% used	Manufacturer lap shear data
6061-T6 / 316 SS / 1020	241 / 205 / 210 MPa yield	Handbook minimums
Nylon 6/6 shear pins	66 MPa shear	Material data
Recovery hardware ratings	8,450 / 5,338 / 3,560 / 4,893 N	Slide 29 assumptions; quick link unresolved (A10)
Parachute coefficients	C_x 1.2 / 1.4, n = 8, X_1–A curve	Knacke, Parachute Recovery Systems
Thermal	200 °C case; k 0.61 W/m·K; c_p 999 J/kg·K	Slide 9 boundary assumption; NIST G-10
Assumed (to confirm by test)	50 g deploy; 86 g av bay; μ 0.10; A_c 15 mm^2; ε 4%; ×2.12	Team assumptions

All figures and diagrams created with the use of AI assistance